



化学测量学—人工智能化学分析

Chapter 2

Machine-Learning Basics

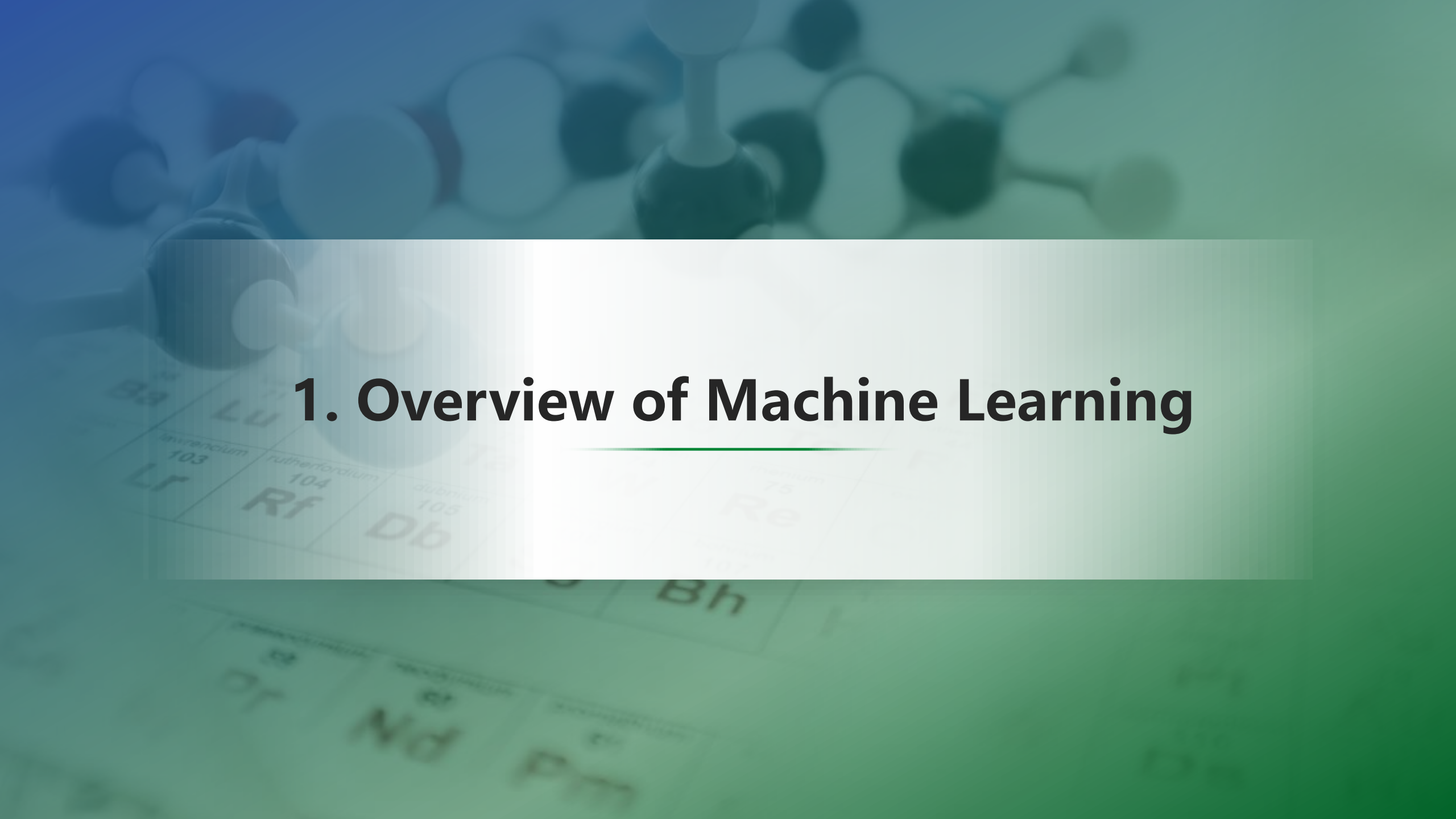
Regression, Classification, Clustering

人工智能化学分析

汪骋 (Wave) 厦门大学

Contents

- 1. Overview of Machine Learning**
- 2. The Key Components of Machine Learning**
- 3. Basic Models of Machine Learning**
- 4. Implementing in Python for Chemistry**



1. Overview of Machine Learning

What is Machine Learning ?

What is learning?

Herbert A. Simon →

Learning is any process by which a system improves performance from experience

Tom M. Mitchell →

Machine Learning is the study of algorithms that improve their performance P at some task T with experience E .



Herbert A. Simon

Nobel Prize

Turing Award

Decision Making



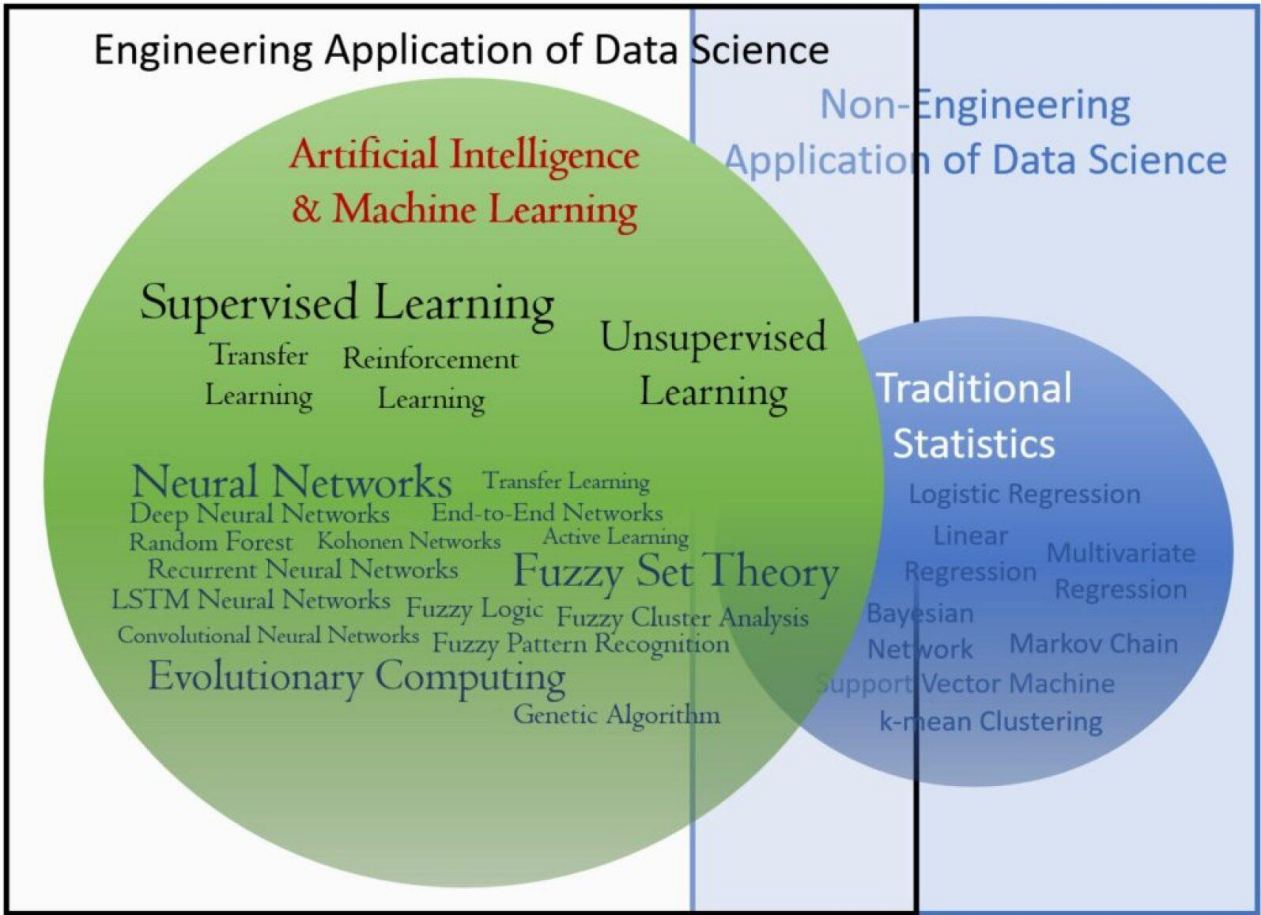
Tom M. Mitchell

American Academy of Arts and Sciences

National Academy of Engineering

Machine Learning and Statistics

Glossary	
Machine learning	Statistics
network, graphs	model
weights	parameters
learning	fitting
generalization	test set performance
supervised learning	regression/classification
unsupervised learning	density estimation, clustering
large grant = \$1,000,000	large grant= \$50,000
nice place to have a meeting: Snowbird, Utah, French Alps	nice place to have a meeting: Las Vegas in August



(Face recognition; handwriting/speech recognition; translation; medical image; Autonomous driving)

(Dall E; ChatGPT; 3D design)

(Time series prediction)

Speech Recognition

Facial Recognition

MACHINE LEARNING

Image Recognition

Autonomous Driving

Optical Character Recognition

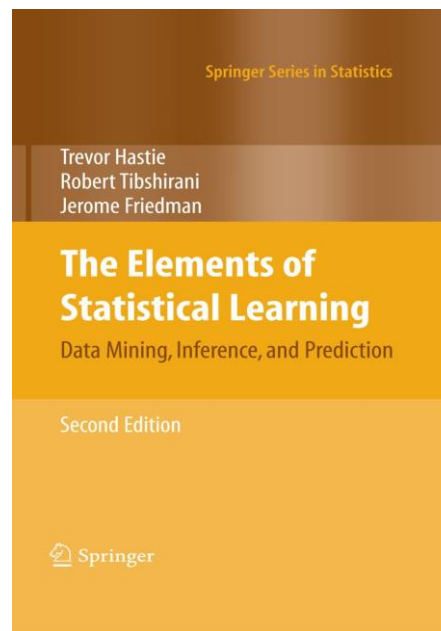
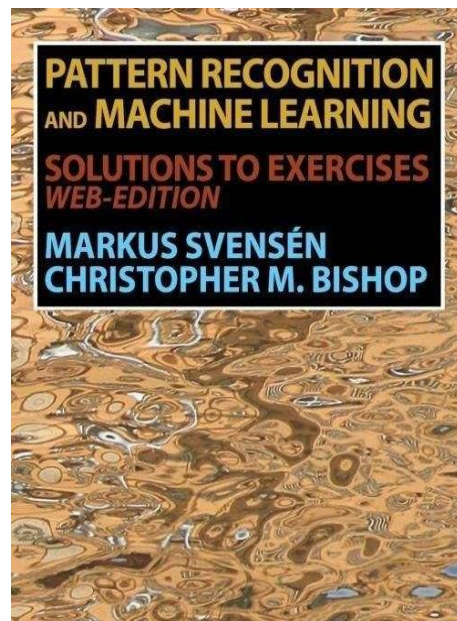
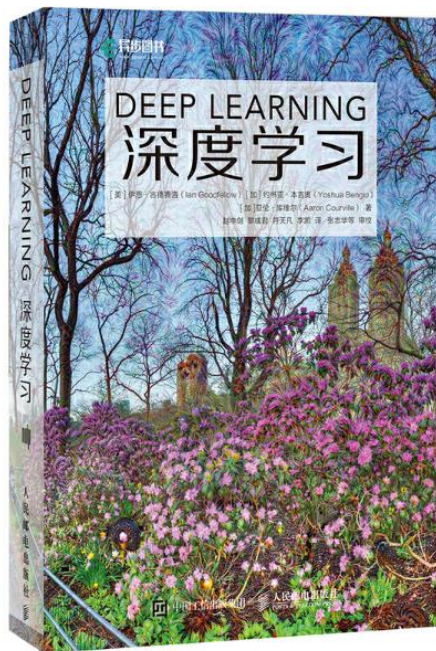
Numerous Resources to Master ML

- <https://www.coursera.org/learn/machine-learning>
(Andrew Ng, entry level)
- <https://davidrosenberg.github.io/ml2019/#home>
(NYU, core course, advanced level)
- <https://tw.coursera.org/learn/ntumlone-mathematicalfoundations>
- <https://www.coursera.org/learn/ntumlone-algorithmicfoundations> (Chinese)

It is possible to quickly become an expert in ML

Text books for Machine Learning ?

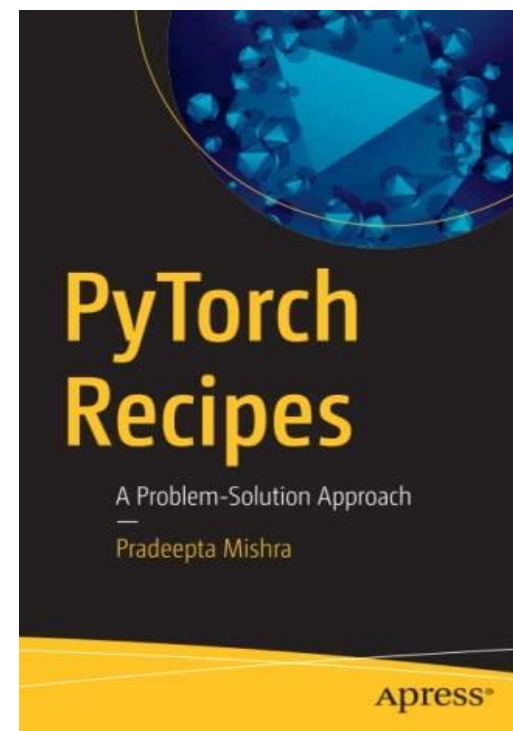
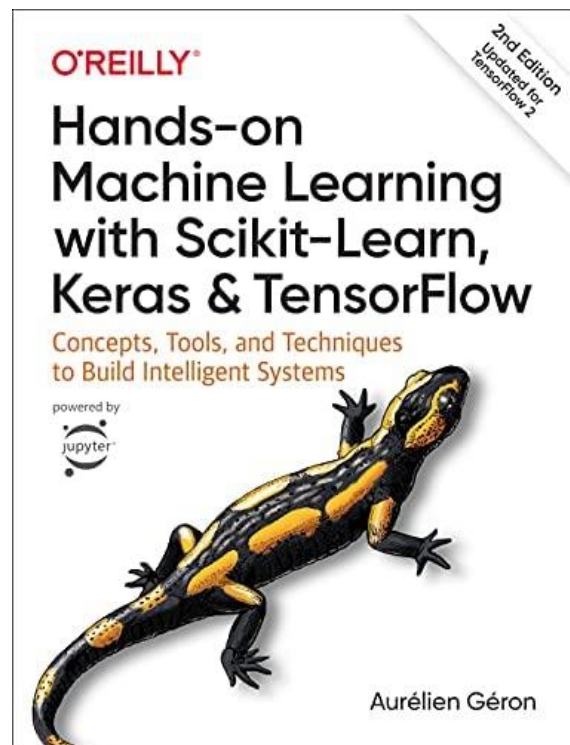
Too many of them. Not recommended for in-depth reading.
Learn through battles!



Text books for Learning from practice!

Still not recommended for most chemistry majors.

Find data in your own lab!!!



Large Language Models

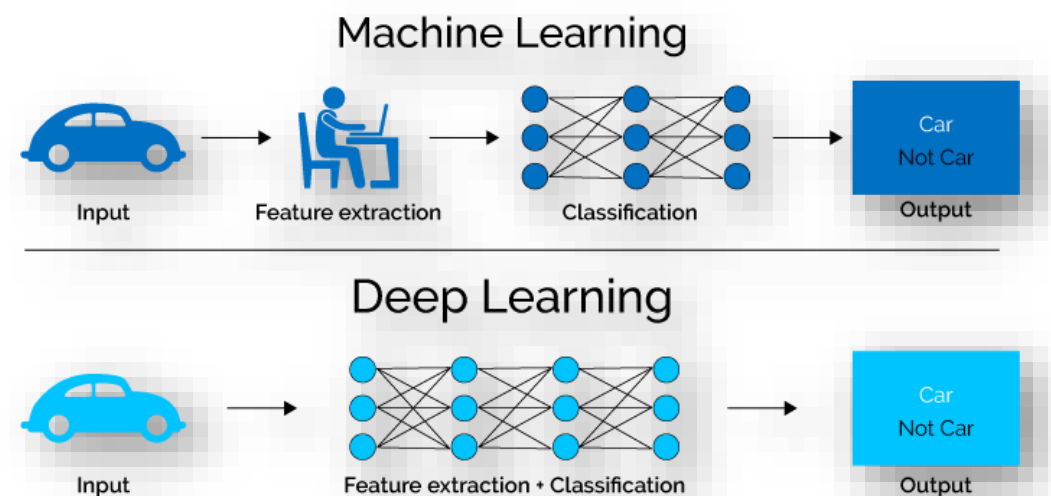
LLMs can help a lot in coding, but make sure to go through the generated codes line by line and tested them thoroughly



Types of Machine Learning

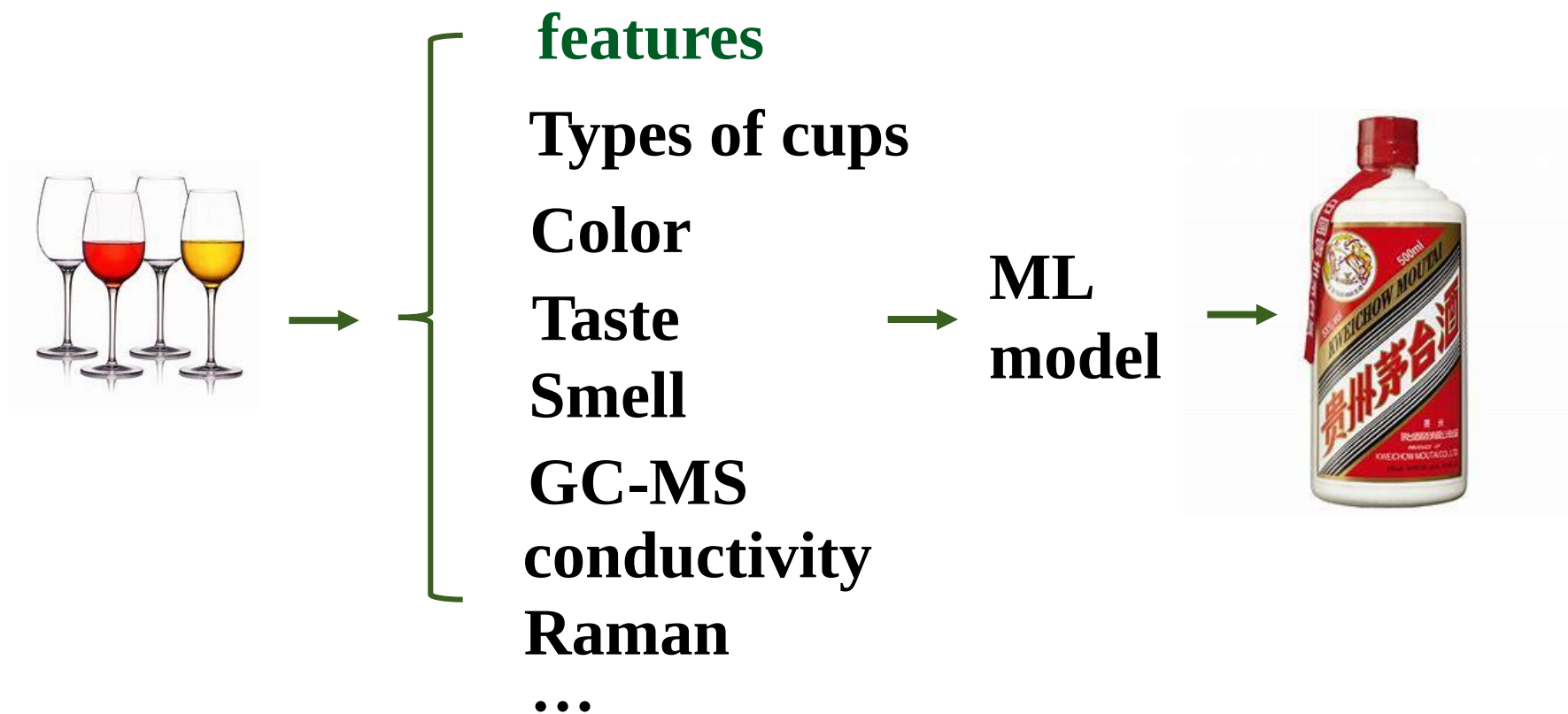
Can the model automatically extract features in the data?

- **Deep learning** : Autonomous feature extraction
- **Feature engineering + machine learning**: human-curated features (domain knowledge of the expert)



A fake example throughout the slides to demonstrate the principles

Is it Moutai?



Types of Machine Learning

Is labeled data (often human curated) needed ?:

➤ Supervised :

Classification

Regression

Pattern Recognition

➤ Unsupervised:

Clustering

Dimension Reduction (feature learning)

Rule Learning

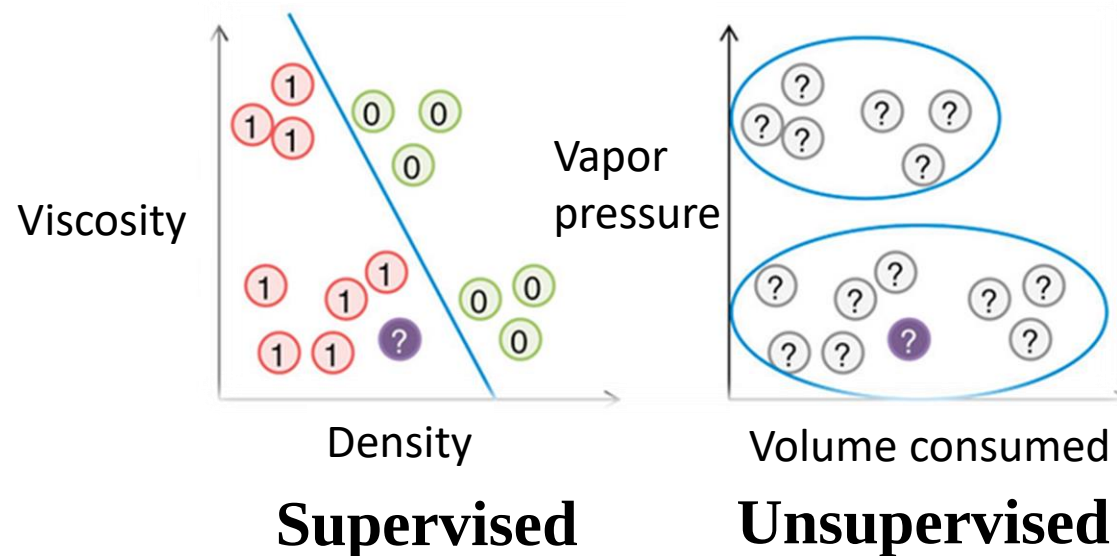
Anomaly Detection

Density Estimation

Autoencoder (learning embedding)

Data Augmentation

Is it Moutai?



Types of Machine Learning

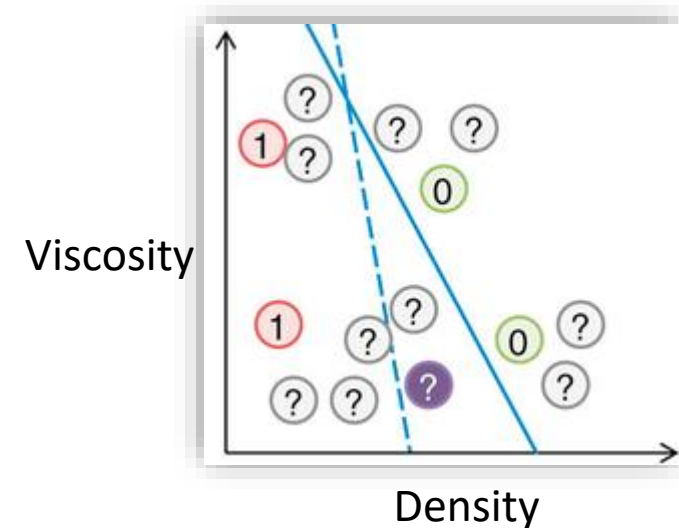
Semi-supervised : Use both labeled data and unlabeled data

Self-training: trained with a small set of labeled data, predicting labels for the unlabeled data for retraining.

Co-training: Two (or more) models are trained separately on the labeled data and then predict on the unlabeled data. The models can then teach each other from their predictions.

Transductive learning: The model tries to predict the labels for the unlabeled data. The focus is not on generalization to unseen data, but rather on predicting the labels of the given unlabeled data.

Active learning



Types of Machine Learning

Self-supervised : Pre-training language models

Design a method to use the context before and/or after a word to predict the right word to fit in.

It does not require labelling, but it is trained like a supervised model.

Types of Machine Learning

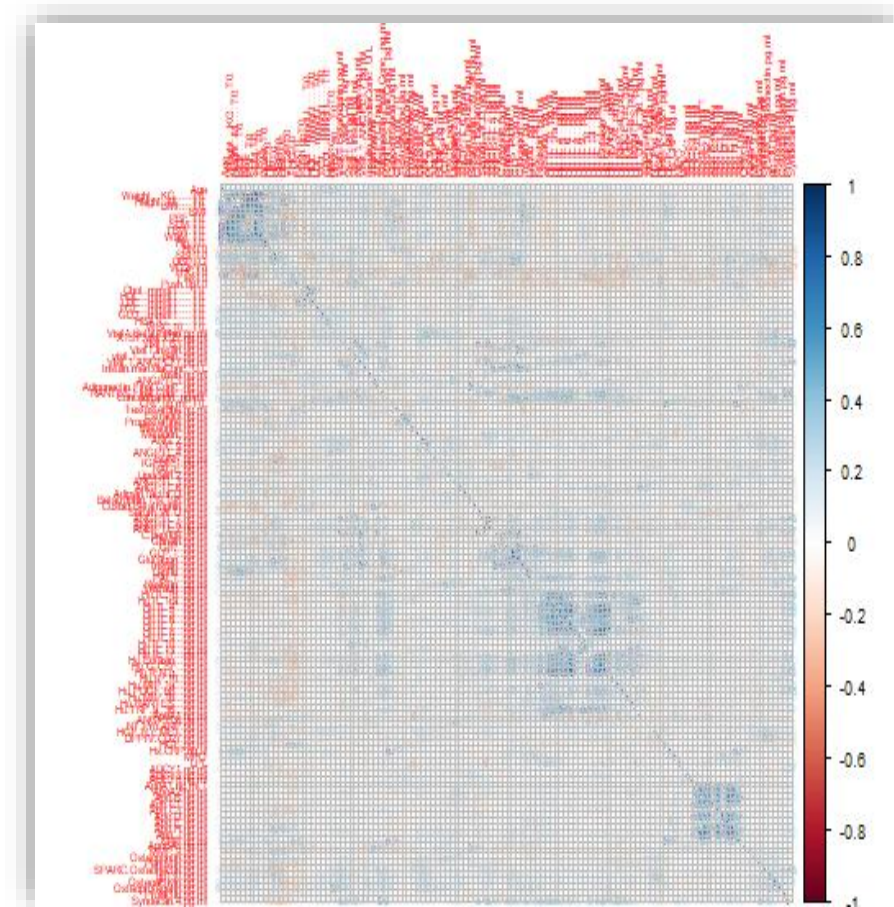
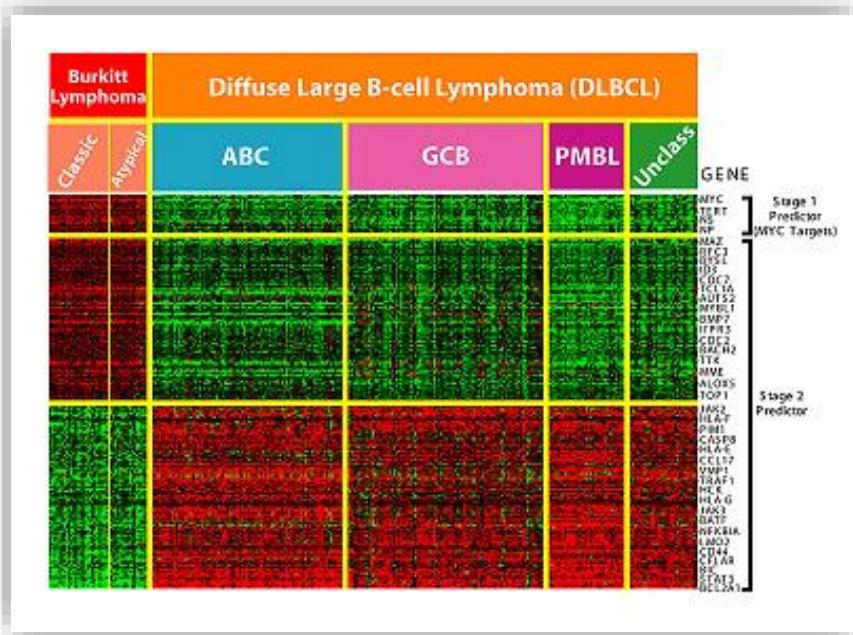
Multiple other terms

- Reinforcement Learning
- Transfer Learning
- Federated Learning
- Online Learning
- Causal Learning
- Active Learning
- Meta Learning
- Few-shot Learning
- Online Learning
- Adversarial Learning

The Key Issue to Solve by ML

Multiple variables, High dimensional data

- Data sparseness
- High cost in calculation



The Curse of Dimensionality

Data Sparsity: As the number of dimensions increases, the volume of the space increases so fast that the available data become sparse. The data needed to support the statistical and machine learning models grows exponentially.

Computational Complexity: Many algorithms that are efficient in low dimensions become intractable when the input is high-dimensional.

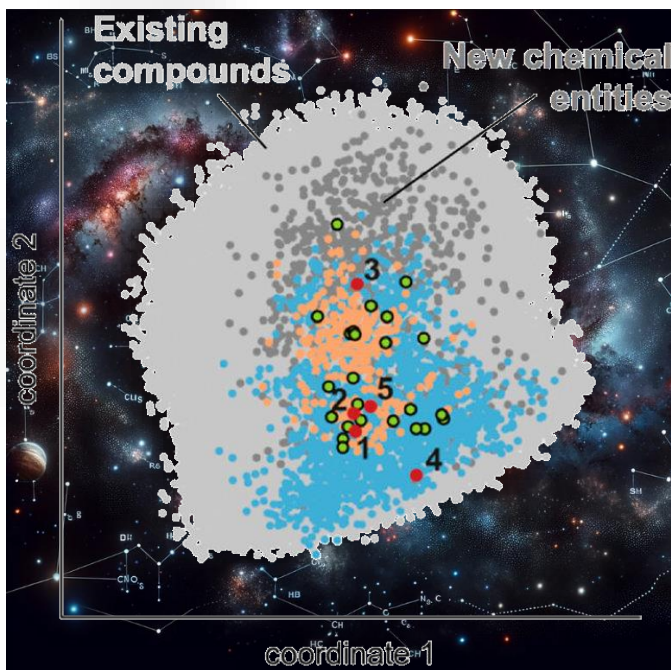
Especially for Chemistry What can we do ?

The Key Issue to Solve by ML

The Chemical Space

Diverse chemicals

10^{60} molecules



+

Complicated Interactions

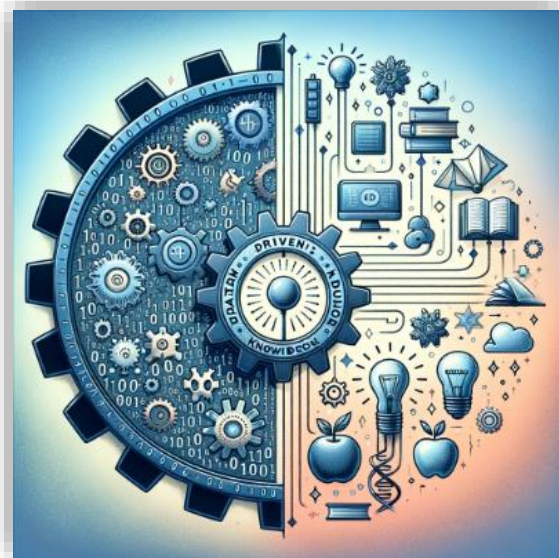
Reaction, assembly, engineering



The Key Issue to Solve by ML

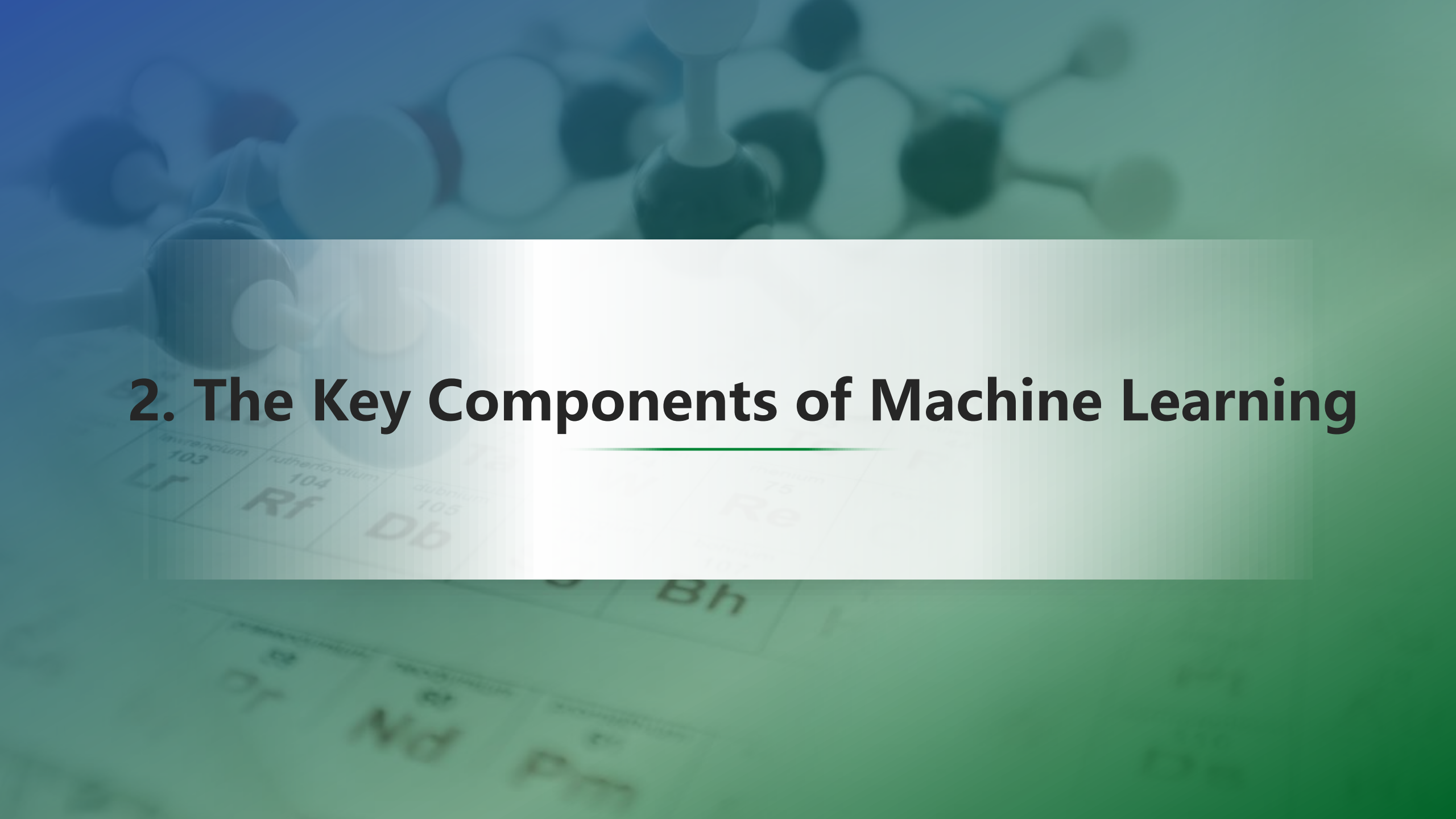
Formulation Science heavily relied on data-driven approach, but even formulation search needs some knowledge guidance

Obtaining a formulation like Moutai by just data-driven ?



One key issue: Data-driven and Knowledge-driven are not combined!

It is unlikely to use pure data-driven strategy to explore such a vast and sparse space.



2. The Key Components of Machine Learning

The Key Components of ML

2.1 Data Preparation

2.2 Models

2.3 Algorithms

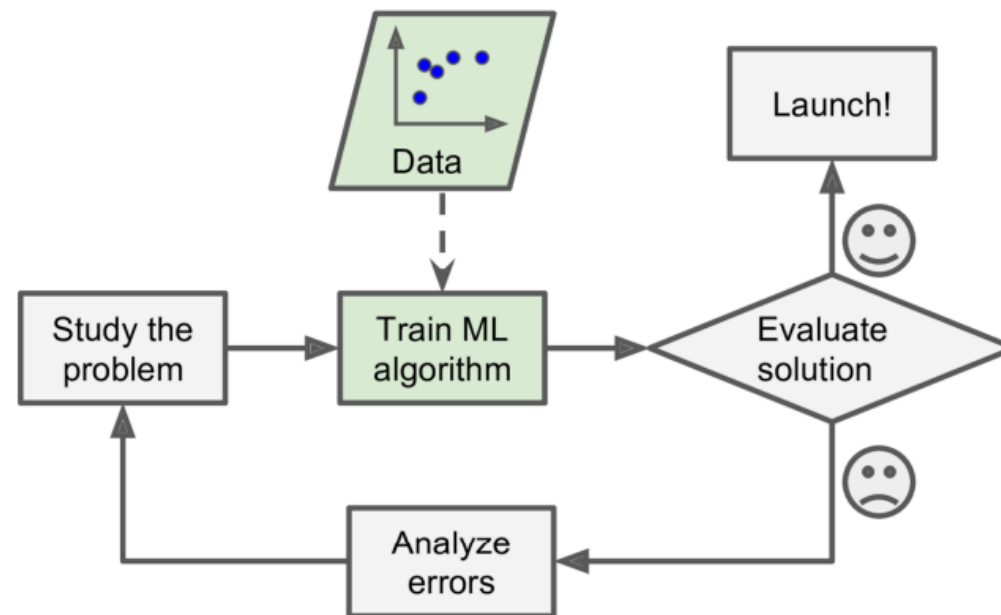
2.4 Model Interpretation

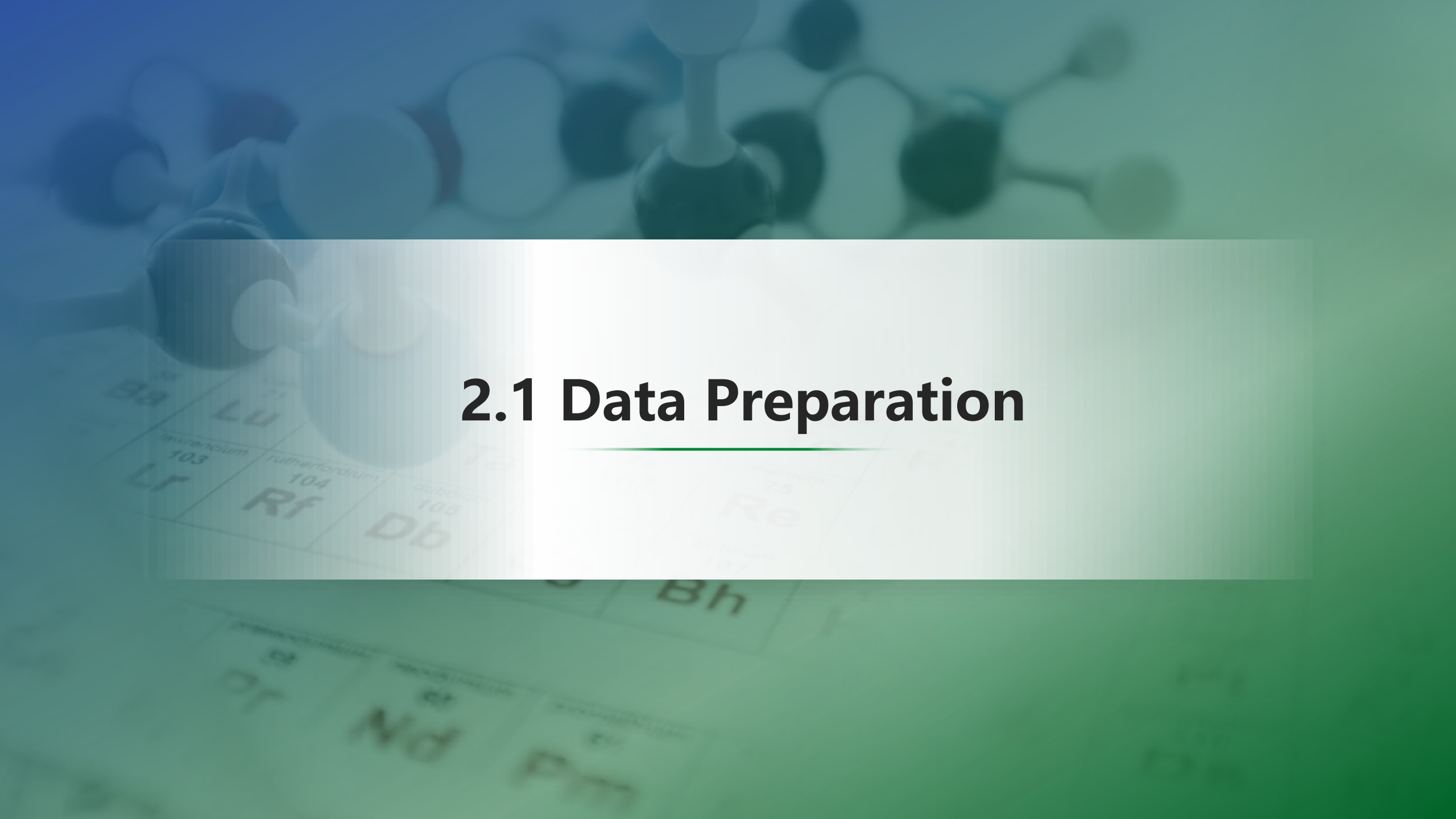
Steps in a Machine Learning

- Data Splitting (Training set, Validation set, Test set)
- Build model (The model, loss function, the algorithm)
- Model Evaluation (Figure of Merit)

Is it Moutai?

Data collection includes featurization





2.1 Data Preparation

Feature Extraction (Data representation)

- How to vectorize data?
- How to reduce vector dimension?

Underlying assumption:

Any task can be finished by considering just a few features

You just need to find the right ones

Is the document describing Moutai?

	aardvark	0
	abacus	0
	...	
	antitrust	42
	...	
	CEO	17
	Baijiu	61
	Wine	14

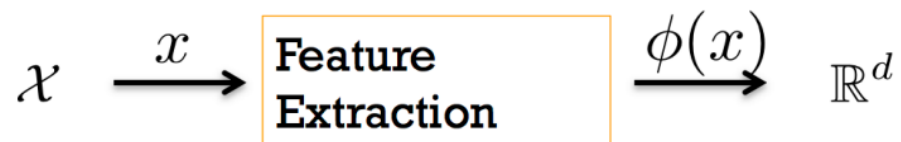
	aardvark	0	2	0
	abacus	0	0	0
	...			
	antitrust	42	0	0
	...			
	CEO	17	0	0
	Baijiu	61	0	0
	Wine	14	0	0

A simple word count representation

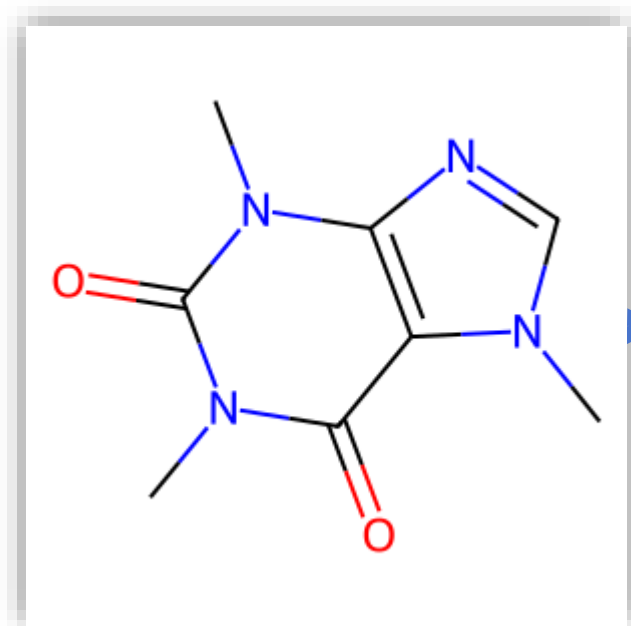
Feature Extraction (Data representation)

Raw Input

Feature Vector



Anything known about the molecule may be used as a feature



Molecular Weight: 194.194

Number of Aromatic Rings: 2

TPSA: 61.82

Number of H-Bond Donors: 0

Number of H-Bond Acceptors: 6

?

?

?

?

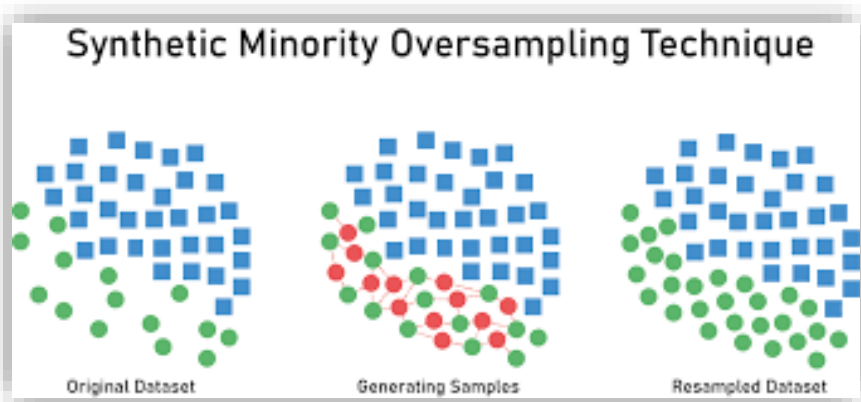
?

Ethanol

Data Augmentation

The purpose of data augmentation is to create a more comprehensive dataset that can represent the variability and diversity of real-world situations the model will encounter. This helps in reducing overfitting and improving the model's ability to generalize from the training data to unseen data.

■ Synthetic data



■ Implement necessary Symmetry

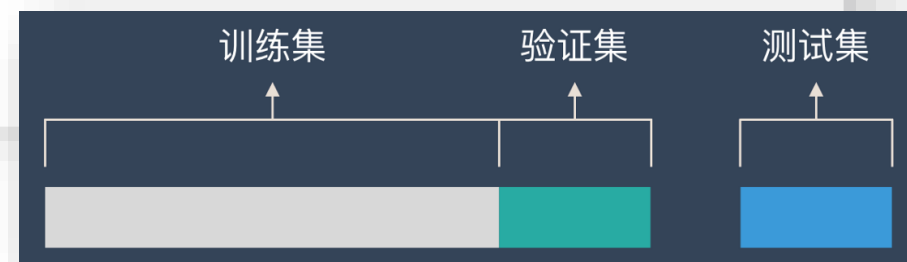


■ Adding noise to the data



Data Splitting

- Training Set (Fitting the parameters in the ML model)
- Validation Set (The hyperparameters)
- Test Set (Model evaluation)



This is non-trivial:

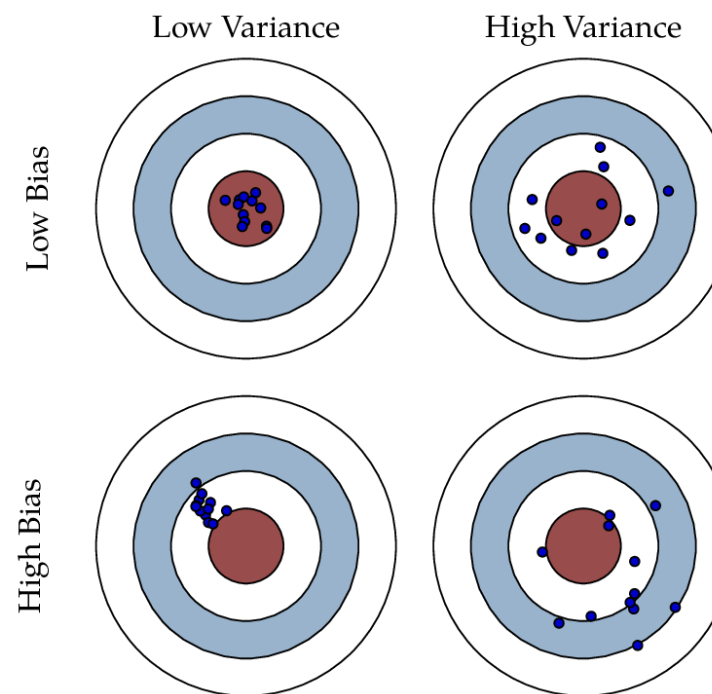
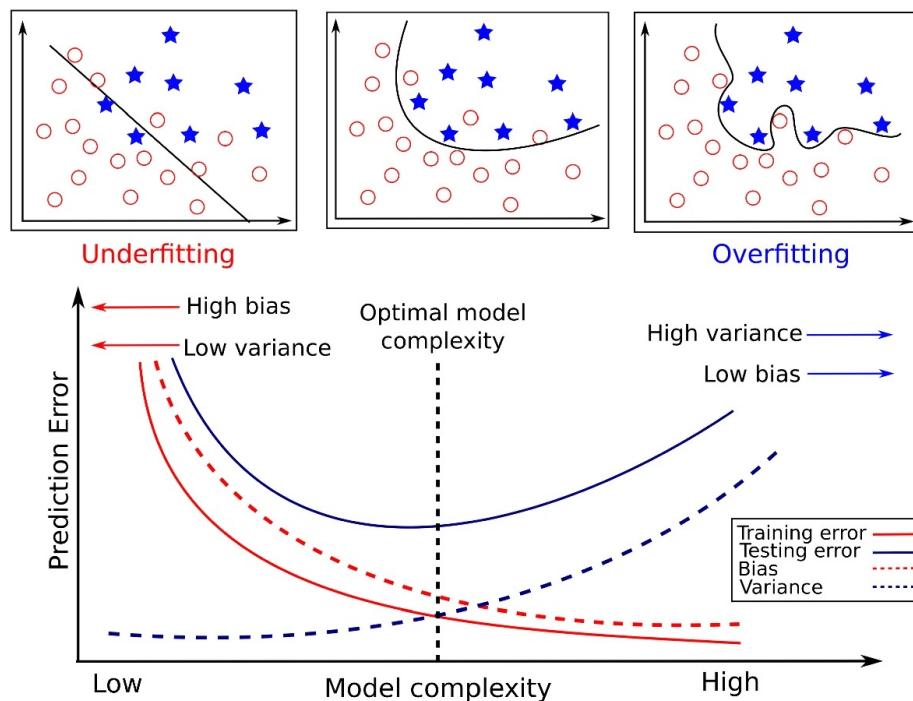
Should we put structurally similar molecules into different sets?

Homoscedasticity: datasets have the same distribution

Homoscedasticity is a basic requirement of the statistics, but if you put similar molecules into train/val/test sets, the high performance may only reflect learning similarity

Overfitting and Underfitting

Bias-variance tradeoff



$$E_{D,\epsilon} \left[(y - \hat{f}(x; D))^2 \right] = \left(\text{Bias}_D [\hat{f}(x; D)] \right)^2 + \text{Var}_D [\hat{f}(x; D)] + \sigma^2$$

Example

Predicting the Redox Potentials of Phenazine Derivatives using DFT Assisted Machine Learning

Siddharth Ghule^{*1,2}, Soumya Ranjan Dash^{1,2}, Sayan Bagchi^{1,2}, Kavita Joshi^{*1,2}, Kumar
Vanka^{*1,2}

¹Physical and Materials Chemistry Division, CSIR-National Chemical Laboratory (CSIR-NCL),

Dr. Homi Bhabha Road, Pashan, Pune 411008, India

²Academy of Scientific and Innovative Research (AcSIR), Ghaziabad 201002, India

Table 1. Some representative structures from training-set/internal test-set

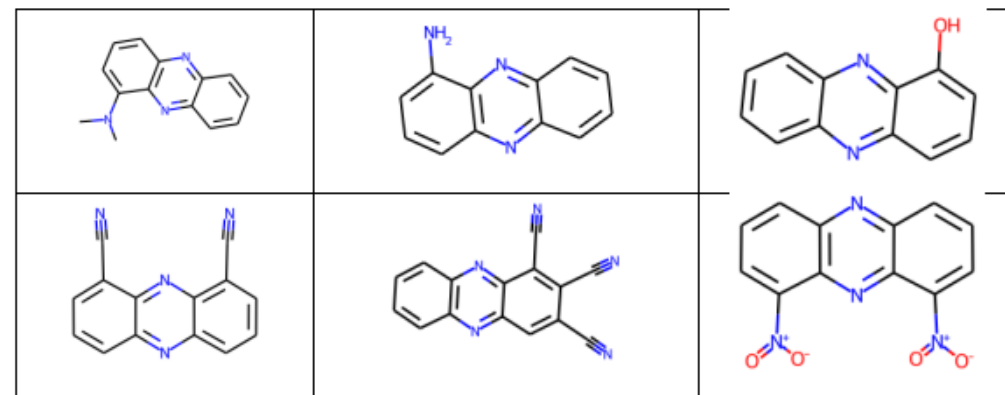
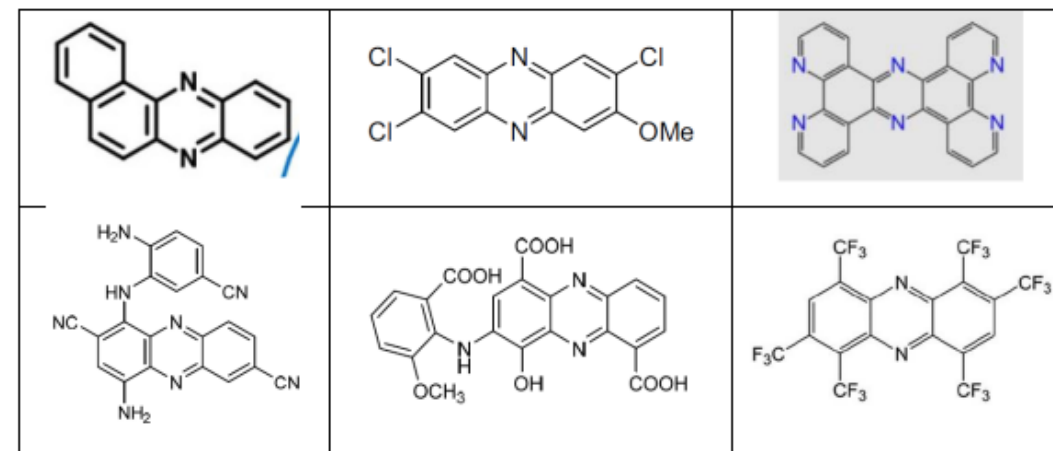


Table 2. Some representative structures from external test-set



Feature Selection

Part of the hyperparameter optimization of the model

Hyperparameters are selected by the performance of **validation sets**

- **Automatic Relevance Determination Regression (ARDR):** This is the probabilistic model related to the sparse Bayesian learning (SBL) framework. It assumes axis-parallel, elliptical Gaussian distribution for each coefficient. The precision of each Gaussian distribution is drawn from the prior distribution (gamma distribution); therefore, it can lead to sparser coefficients. Thus, it is an effective tool to remove irrelevant features^{51,52}.

Model Performance

Validation Set

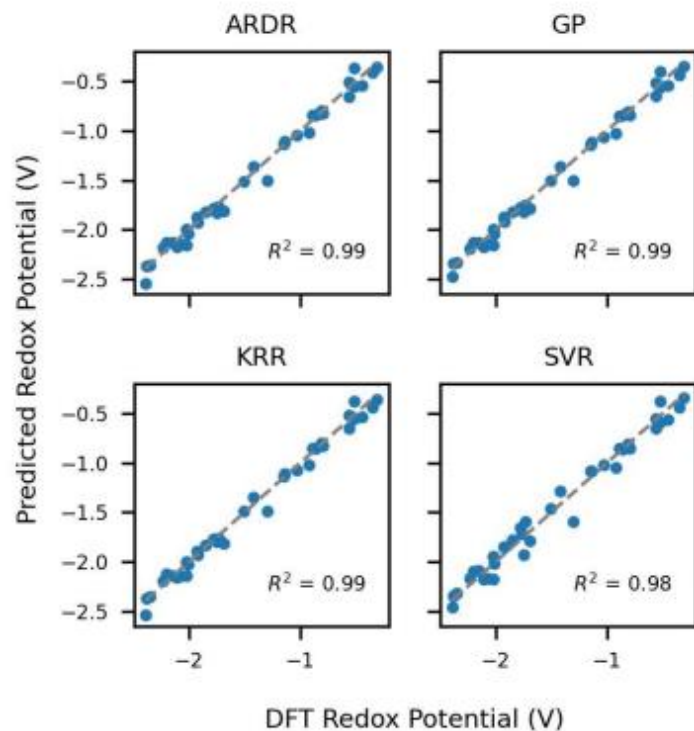


Figure 2. Plots showing machine learning predictions on internal test-set (y-axis) vs. DFT redox potentials (x-axis). Gray dash line corresponds to the perfect predictions.

Test Set

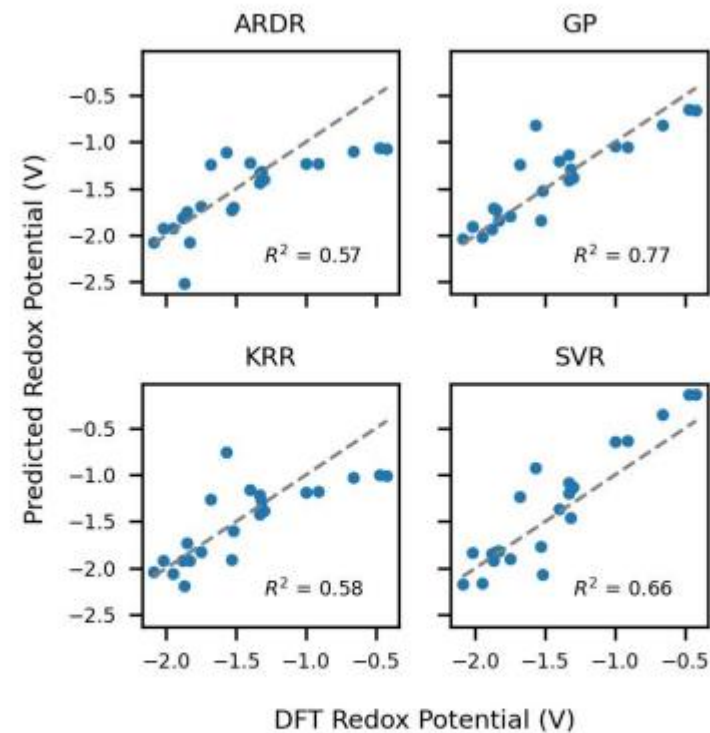


Figure 3. Plots showing machine learning predictions on external test-set (y-axis) vs. DFT redox potentials (x-axis). Gray dash line corresponds to the perfect predictions.

Feature Selection

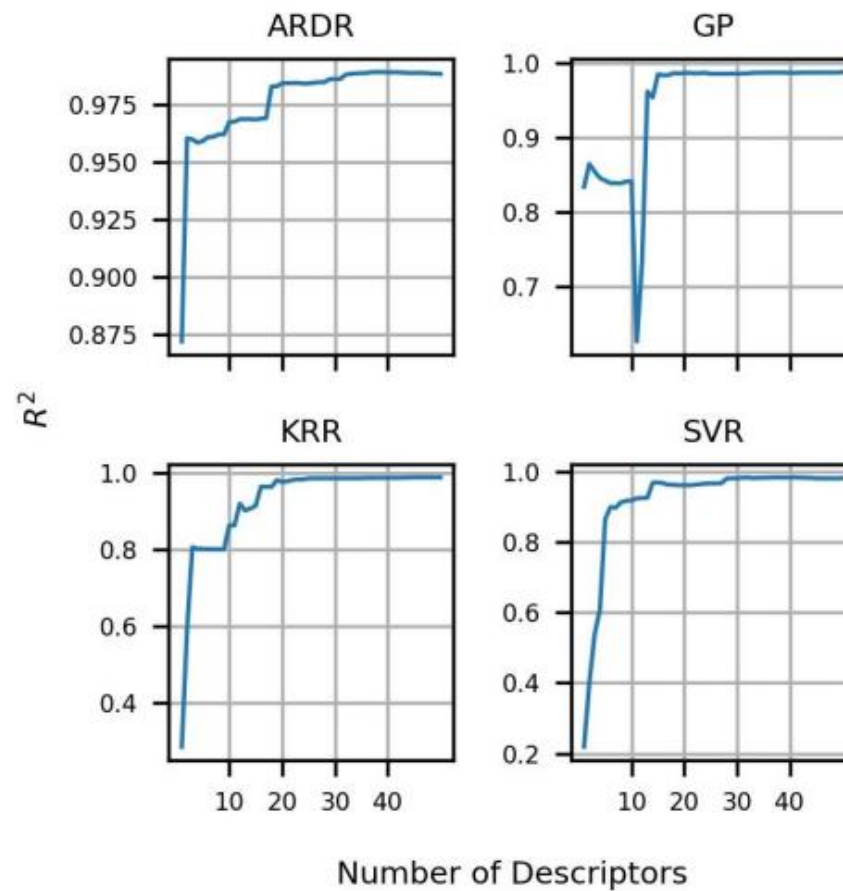
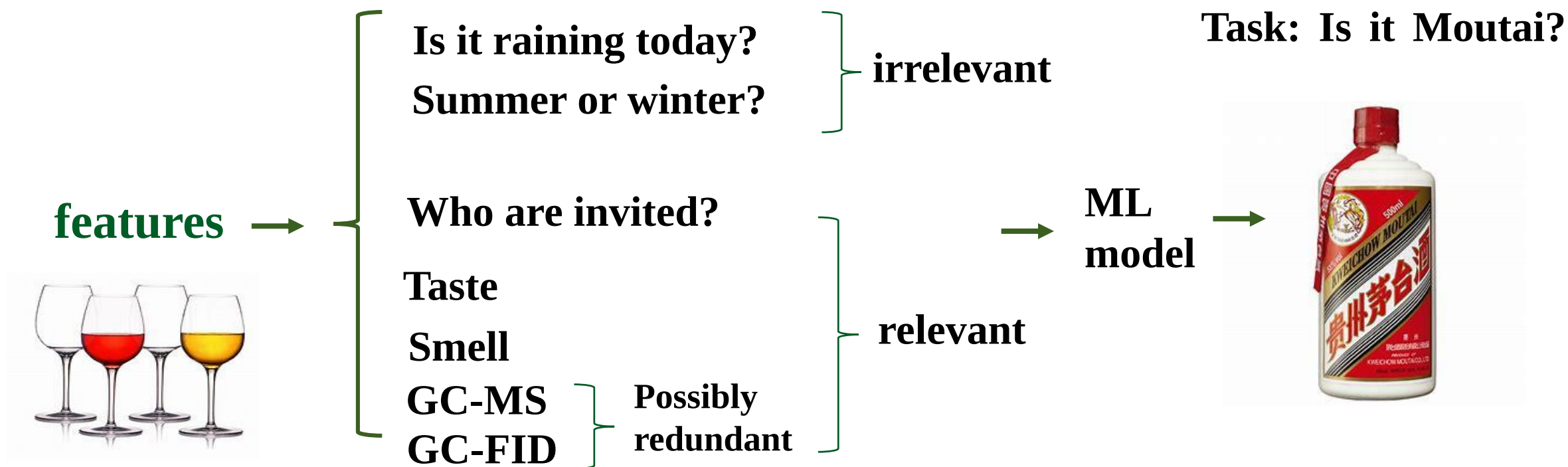


Figure 7. R^2 vs. number of descriptors. R^2 was computed using the internal test-set.

Features

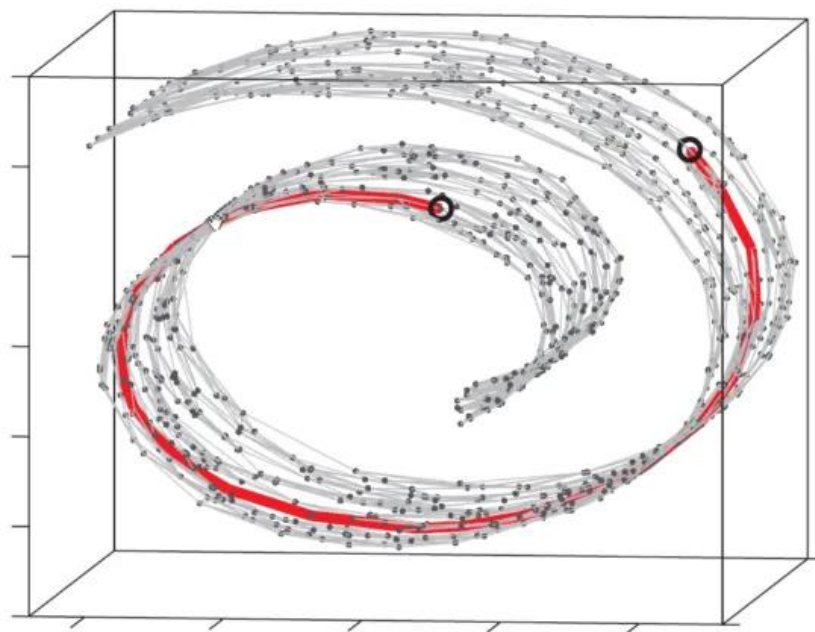
- **Relevant Features:** Features that are beneficial to describe the task
- **Irrelevant Features:** Features that are unrelated to the task
- **Redundant Features:** Features that can be derived from other features



Why dimension reduction can work?

There are always excess information

High-dimensional data can be mapped to a low-dimensional manifold that is more relevant



Feature Reduction

There are also unsupervised ways to reduce the dimensionality of the data

Embedding

A low dimensional “**embedding**” of a high dimensional data may be enough for a certain task.

The dimensionality of many data sets is only artificially high; although the data points might exist in a high-dimensional space, they actually lie on a low-dimensional **manifold** within that space. The real-world high-dimensional data lie on low-dimensional manifolds embedded within the high-dimensional space.

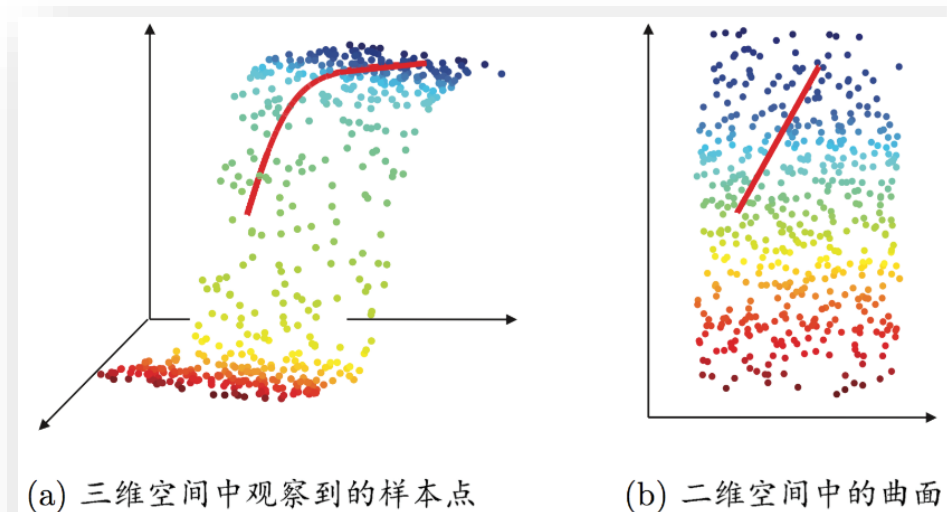


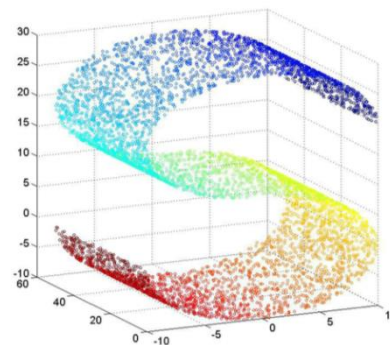
图 10.2 低维嵌入示意图

Dimensionality Reduction Techniques

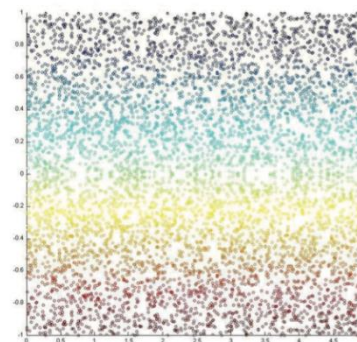
Principal Component Analysis (PCA): Linear dimensionality reduction while retaining most of the variation in the data

Kernel-PCA: Getting to nonlinear transformation

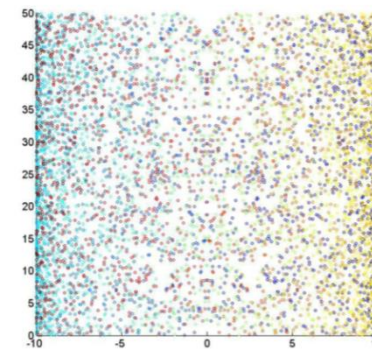
Factor Analysis: Group and transform related features together



(a) 三维空间中的观察



(b) 本真二维结构



(c) PCA 降维结果

图 10.6 三维空间中观察到的 3000 个样本点, 是从本真二维空间中矩形区域采样后以 S 形曲面嵌入, 此情形下线性降维会丢失低维结构. 图中数据点的染色显示出低维空间的结构.

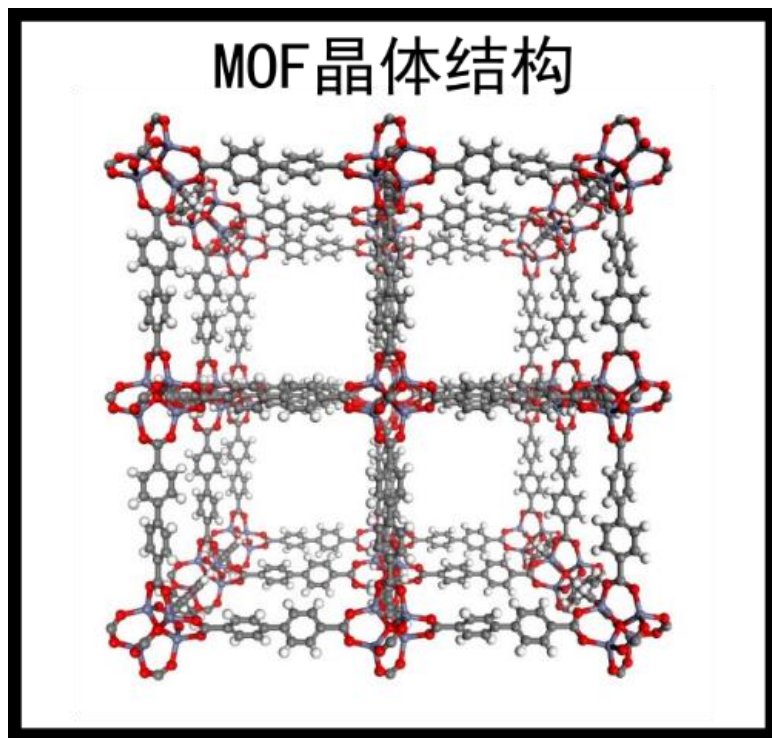
Dimensionality Reduction Techniques

Manifold Learning:

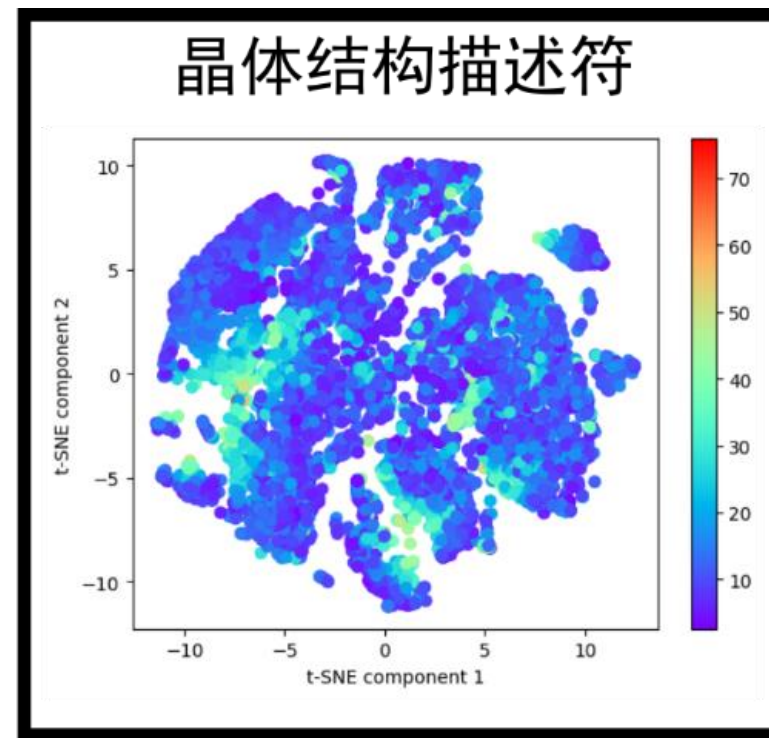
- **Isomap (Isometric Mapping):** Seeks a lower-dimensional embedding which maintains geodesic distances between all points.
- **Locally Linear Embedding (LLE):** Computes a low-dimensional projection of the data which preserves distances within local neighborhoods.
- **t-Distributed Stochastic Neighbor Embedding (t-SNE):** Converts the similarity between data points to joint probabilities and tries to minimize the Kullback–Leibler divergence between the joint probabilities of the low-dimensional embedding and the high-dimensional data.
- **Uniform Manifold Approximation and Projection (UMAP):** A manifold learning technique that is particularly effective at preserving both local and more global structure in the data.

Dimensionality Reduction Techniques

t-SNE for visualization



特征化
→

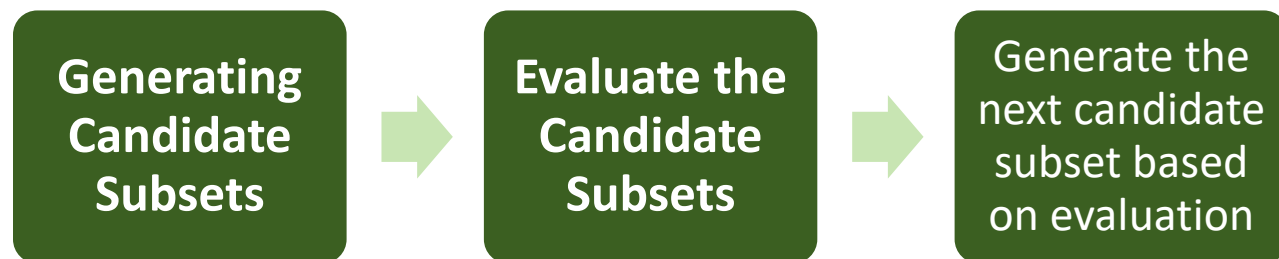


Approach for Feature Selection

Screening for all subsets

Combination Explosion, **impractical**

Practically



Key steps: Subset search and subset evaluation

Subset Search

Using Greedy Algorithm

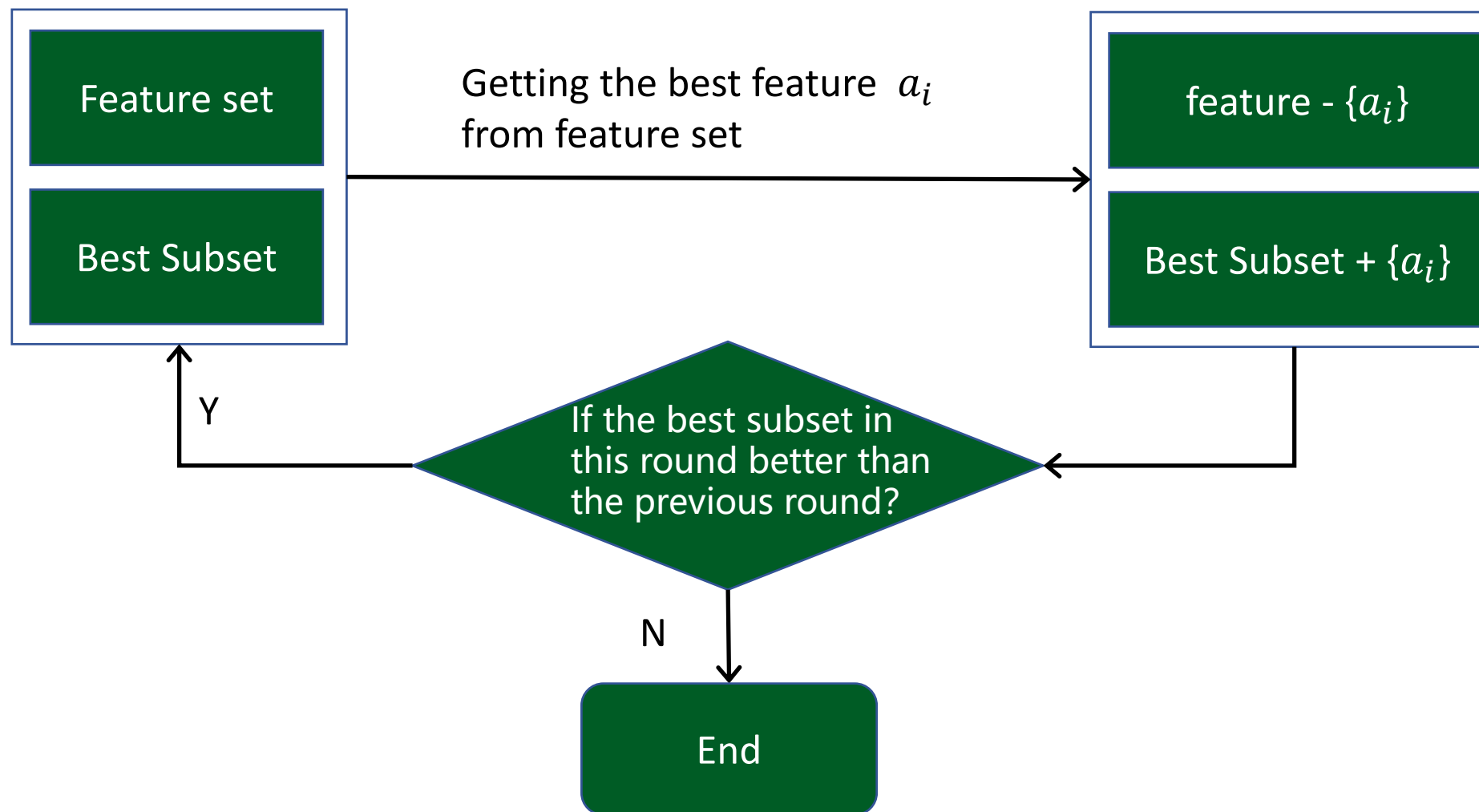
A Greedy Algorithm is a straightforward, intuitive algorithmic paradigm that makes locally optimal choices at each stage with the hope of finding a global optimum.

Forward Search: Increase Features from zero

Backward Search: Eliminate Features from a Full Matrix

Bi-directional Search: Increase Features Each Time and also eliminate Irrelevant Features

Forward Search



Subset Evaluation

The subset values can define a division of the data

The target values can define a division of the data

**The more the two divisions match each other,
the better the subset selection**

Feature Selection Methods

1. Filter Methods:

Statistical Tests (Relief): Using tests like the chi-squared test, ANOVA, or correlation coefficients.

How to determine which is the relevant statistical test?

Relief for classification

Variance Threshold: Removing features whose variance does not meet some threshold.

Feature Selection Methods

2. Wrapper Methods (using model performance):

Recursive Feature Elimination (RFE): Recursively removing features based on feature importance.

Forward Selection: Starting with no features and adding one feature at a time based on val performance.

Backward Elimination: Starting with all features and removing one at a time based on val performance.

Las Vegas Wrapper: Randomly selecting subset and evaluate the performance for multiple times

Feature Selection Methods

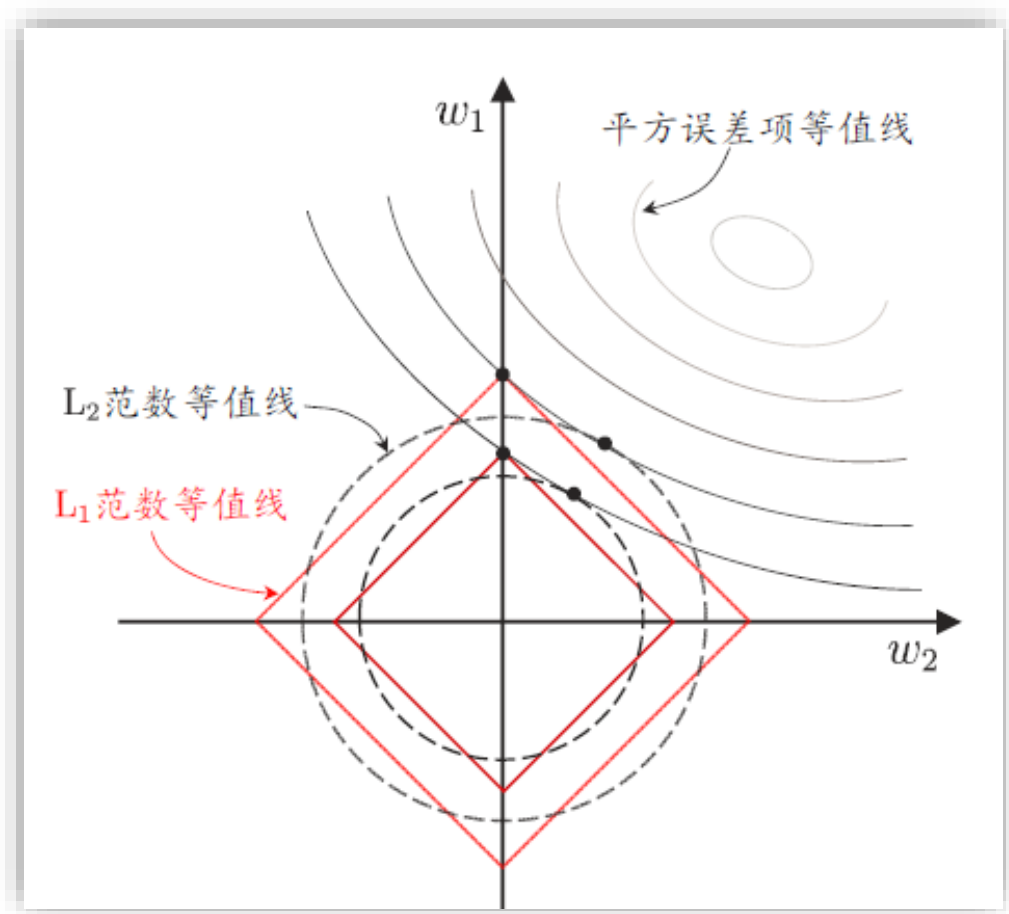
3. Embedded Methods:

LASSO (Least Absolute Shrinkage and Selection Operator):

Linear model with a penalty term which can shrink some coefficients to zero.

Decision Trees: Inherently perform feature selection by selecting the most informative ones for splitting.

Feature Selection Methods



L2 regularization: adding square term of regression coefficient in loss function

Ridge regression

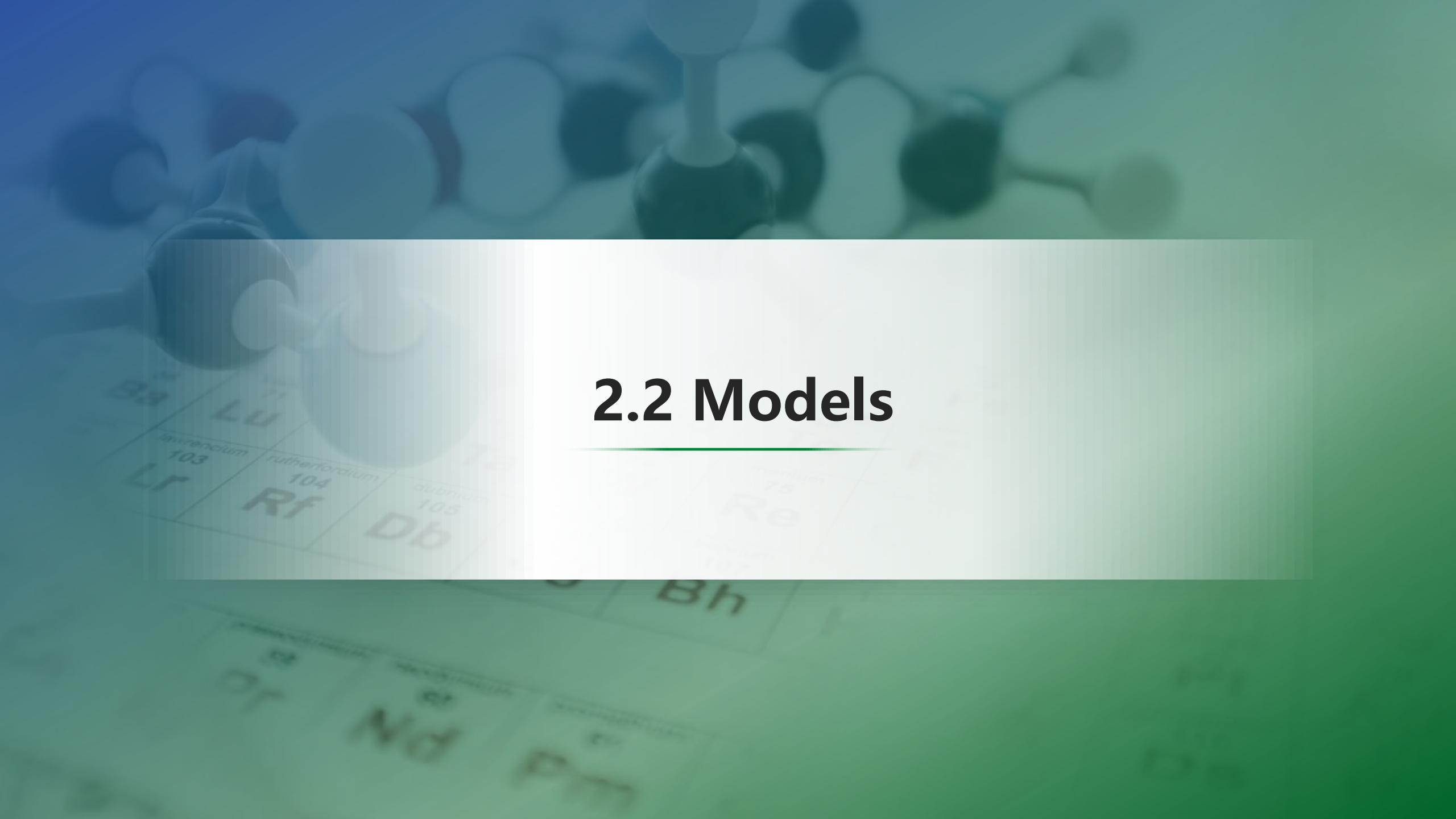
$$\min_{\mathbf{w}} \sum_{i=1}^m (y_i - \mathbf{w}^\top \mathbf{x}_i)^2 + \lambda \|\mathbf{w}\|_2^2$$

L1 regularization: adding absolute term of regression coefficient in loss function

LASSO

$$\min_{\mathbf{w}} \sum_{i=1}^m (y_i - \mathbf{w}^\top \mathbf{x}_i)^2 + \lambda \|\mathbf{w}\|_1$$

Get sparse solution

The background of the slide features a blurred image of a molecular model with white and grey spheres connected by rods, and a portion of a periodic table. The periodic table shows elements like Lu, Rf, Db, Re, Bh, Nd, and Pt. A semi-transparent white box is centered over the image, containing the section header.

2.2 Models

Models

➤ Supervised Learning Models (For classification or regression):

- **Linear Regression:** Predicts a continuous target variable based on input features.
- **Logistic Regression:** Predicts the probability of a binary outcome.
- **Decision Trees:** Makes decisions based on asking a series of questions about the input features.
- **Random Forest:** A collection of decision trees that aggregates their outputs.
- **Support Vector Machines (SVM):** Finds the hyperplane that best separates the classes in the input feature space.
- **Naive Bayes:** Predicts the probability of each class given the input features using Bayes' theorem.
- **K-Nearest Neighbors (KNN):** Classifies a data point based on the majority class among its K nearest neighbors.
- **Gradient Boosting Machines (e.g., XGBoost, LightGBM):** Builds a series of weak learners, typically decision trees, that sequentially correct the mistakes of the previous one.

Models

➤ Unsupervised Learning Models (for clustering):

- **K-Means Clustering:** Groups data into K number of clusters.
- **Hierarchical Clustering:** Creates a tree of clusters.
- **Principal Component Analysis (PCA):** Reduces the dimensionality of the data by projecting it onto a smaller subspace.
- **Autoencoders:** Neural networks that learn to compress and decompress the data in an unsupervised manner.

Models

➤ Neural Networks and Deep Learning Models (classification or regression):

- **Multilayer Perceptrons (MLP)**: A simple neural network with one or more hidden layers.
- **Convolutional Neural Networks (CNN)**: Particularly suited for image recognition tasks.
- **Recurrent Neural Networks (RNN)**: Suitable for sequential data like time series or text.
- **Long Short-Term Memory (LSTM)**: A type of RNN capable of learning long-term dependencies.
- **Generative Adversarial Networks (GAN)**: A pair of networks, one generating data and the other evaluating it.
- **Graph Neural Network (GNN)**: A neural network based on graph data
- **Transformer**: Attention is all you need

Loss Function

Need a figure-of-merit for the algorithm to decide which model parameters are better, quantifying how well the predictions of a model align with the true data values.

The loss function measures the discrepancy between the actual values and the values predicted by the model.

Types of Loss Functions:

- **Mean Squared Error (MSE):** Commonly used in regression tasks, it calculates the average of the squared differences between the actual and predicted values.
- **Mean Absolute Error (MAE):** Calculates the average of the absolute differences between the actual and predicted values.
- **Cross-Entropy Loss (Log Loss):** Widely used in classification tasks, it quantifies the difference between the actual probability distribution and the predicted probability distribution.
- **Hinge Loss:** Often used with Support Vector Machines, it measures the error for classification tasks.

• Cross-Entropy Loss (Log Loss)

Both are differentiable and robust to outlier

• Formula:

- Binary Classification: $L(y, p) = -[y \log(p) + (1 - y) \log(1 - p)]$
- Multi-Class Classification: $L(y, p) = -\sum_i y_i \log(p_i)$

p_i is the predicted probability of one class, y_i is the label

High penalty on misclassification

• Hinge Loss

Usage: Often used in binary classification tasks, particularly with Support Vector Machines (SVMs). It maximizes the margin between classes.

• **Formula:** $L(y, f(x)) = \max(0, 1 - y \cdot f(x))$

In this formula, y is the true label, assumed to be either +1 or -1, and $f(x)$ is the predicted score for the positive class.

Loss Function

Optimization

Gradient Descent:

- Loss functions are typically optimized using gradient descent or its variants, where the gradients of the loss function with respect to the model parameters are computed to update the parameters in a way that minimizes the loss.
- Differentiable loss function can facilitate the optimization

Backpropagation:

- In neural networks, the backpropagation algorithm is used to minimize the loss function by adjusting the weights of the network in response to the error of its predictions.

Loss Function

Regularization:

Sometimes, a regularization term is added to the loss function to prevent overfitting, which penalizes complex models.

L1 Regularization :

- L1 regularization adds the absolute value of the magnitude of coefficient values (L1 norm) to the loss function (LASSO):

$$L_{\text{lasso}}(w) = L(w) + \alpha \sum_{i=1}^n |w_i|$$

It can lead to sparse models where some feature weights can be exactly zero, effectively excluding certain features.

L2 Regularization (Ridge):

- L2 regularization adds the squared value of the magnitude of coefficient values (L2 norm) to the loss function:

$$L_{\text{ridge}}(w) = L(w) + \alpha \sum_{i=1}^n w_i^2$$

It tends to shrink the coefficients but keeps them distributed more evenly across all input features.

Hyperparameter

$$L_{\text{lasso}}(w) = L(w) + \alpha \sum_{i=1}^n |w_i|$$

These are **hyperparameters** that are external parameters defining the model structure

$$L_{\text{ridge}}(w) = L(w) + \alpha \sum_{i=1}^n w_i^2$$

If you want to tune hyperparameters, use the validation set performance

Never use test set performance to tune hyperparameters!!!

That will hide overfitting!!!

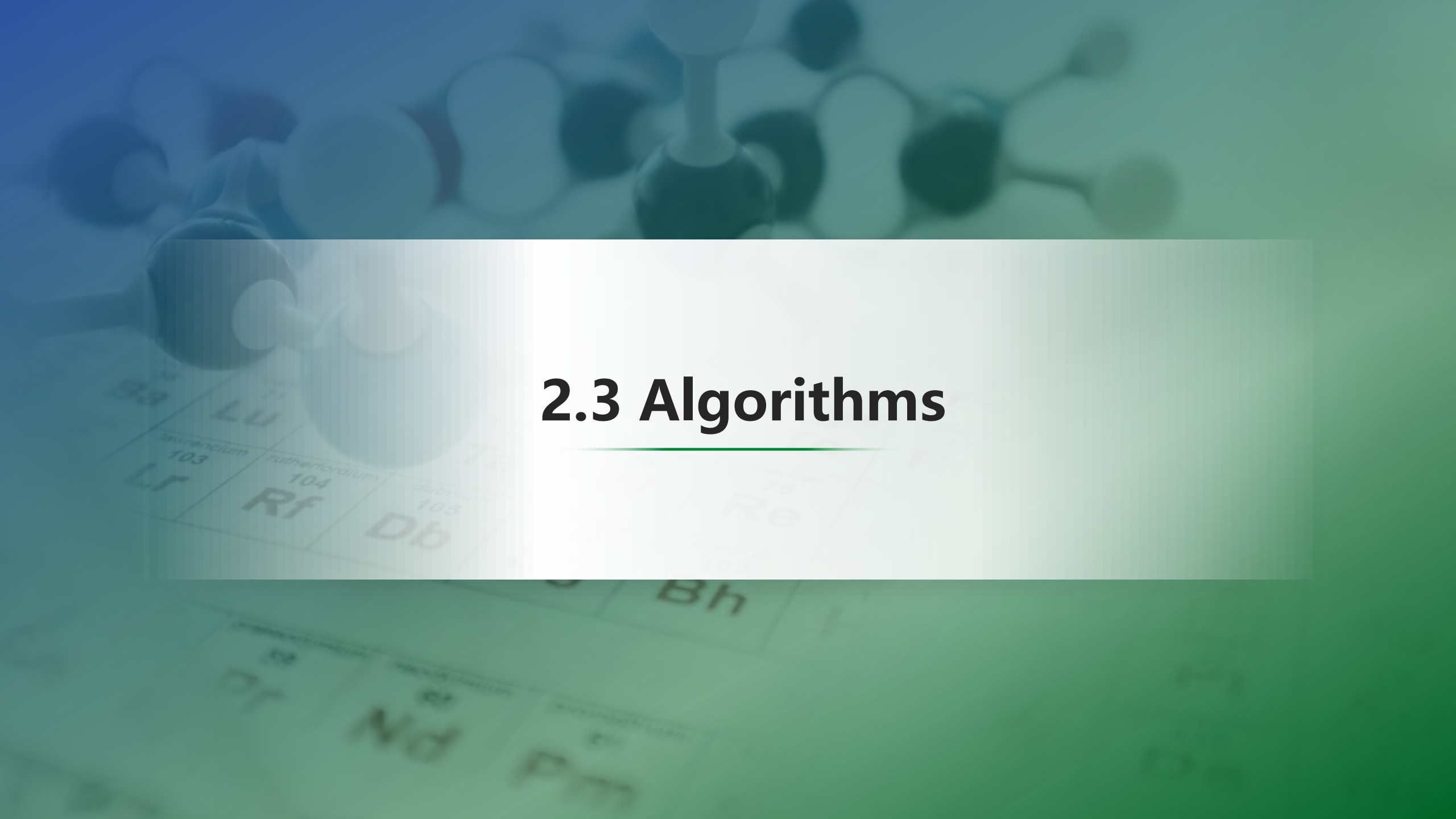
Hyperparameter

Common hyperparameters:

- **Learning Rate:** The step size taken during optimization (e.g., gradient descent).
- **Regularization Term:** Coefficient for L1/L2 regularization terms to prevent overfitting.
- **Number of Clusters** in K-Means clustering.
- **Maximum Depth** of a decision tree or random forest.
- **Number of Hidden Units or Layers** in a neural network.
- **Feature Selection**

Optimization:

Techniques such as grid search, random search, or more advanced methods like Bayesian optimization can be used to find optimal hyperparameters.



2.3 Algorithms

Algorithm

Objective:

Using limited resources (time + hardware) to obtain a reasonable parameter sets of a model based on the training dataset

➤ Gradient Descent (GD):

A basic optimization algorithm that iteratively adjusts the model's parameters in the opposite direction of the gradient of the loss function.

➤ Stochastic Gradient Descent (SGD):

A variant of gradient descent that computes the gradient and updates the parameters using a single training example, which can be faster and capable of handling large datasets.

➤ Mini-Batch Gradient Descent (MB-GD):

A compromise between GD and SGD that computes the gradient using a small batch of training examples.

Algorithm

➤ **Momentum:**

Accelerates SGD by navigating along the relevant directions and dampening the oscillations, making use of a momentum term that considers the previous gradient.

➤ **Adagrad:**

Adapts the learning rates during training, allowing for lower learning rates for parameters associated with frequently occurring features and higher learning rates for parameters associated with infrequent features.

➤ **Adam (Adaptive Moment Estimation):**

Combines ideas from both Momentum and RMSprop, keeping running averages of both gradients and their squares to adapt the learning rate during training.

Simple linear regression

Model

Model Representation:

- The linear regression model can be represented as:

$$y = \beta_0 + \beta_1 x + \epsilon$$

where:

- y is the dependent variable,
- x is the independent variable,
- β_0 and β_1 are the parameters of the model,
- ϵ is the error term.

Loss function

The common choice is the Mean Squared Error (MSE):

$$J(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

Simple linear regression

Algorithm

Gradient Descent:

- Use Gradient Descent to minimize the cost function and find the optimal values of β_0 and β_1 .
- Update rules for β_0 and β_1 in each iteration:
$$\beta_0 := \beta_0 - \alpha \frac{\partial}{\partial \beta_0} J(\beta_0, \beta_1)$$
$$\beta_1 := \beta_1 - \alpha \frac{\partial}{\partial \beta_1} J(\beta_0, \beta_1)$$
where:
 - α is the learning rate.

Analytical Solution (Normal Equation): Least Squares Method

- Alternatively, an analytical solution can be found using the Normal Equation:
$$\beta = (X^T X)^{-1} X^T y$$
where:
 - β is the vector of parameters,
 - X is the matrix of input features (with a column of ones for the intercept term),
 - y is the vector of target values.

Model + Loss Function + Algorithm = Method

	Model	Loss Function	Algorithm
Classical Linear Regression	Linear Model	MSE	Gradient descent or analytical
Lasso	Linear Model	MSE + L1 Regularization	Coordinate descent or subgradient descent
Ridge Regression	Linear Model	MSE + L2 Regularization	Gradient descent or analytical

Model Evaluation

- Using test set for the Evaluation
- Using validation set for the hyperparameter selection
- The loss functions for the training and evaluation are not necessarily the same

Cross-Validation:

- Hyperparameters are often selected through cross-validation, where the training set was divided into K parts. Every time, one part is used for validation and the other $(K-1)$ parts are used for training. The model performance is evaluated on the average of the N -fold cross-validation. In this case, there is not explicit split of the dataset to contain a validation set, but the validation set is adopted.
- **Leave-One-Out Cross-Validation (LOOCV)**: A special case of K -fold cross-validation where K equals the number of data points. It's computationally expensive but can be useful for very small datasets.
- For small dataset, it is necessary to split the dataset for multiple times to evaluate the performance to avoid wrong estimation.

Model Evaluation

Bootstrap Resampling:

- Randomly drawing with replacement from the dataset to create bootstrap samples on which the model is trained and evaluated. Each observation can be chosen more than once in each new sample.

Repeated Training-Test Splits:

- Randomly splitting the data into training and testing sets multiple times and averaging the results.

Model Evaluation

Classification Problems:

- **Accuracy:** The ratio of correctly predicted instances to the total instances.
- **Precision, Recall, and F1-Score:** Precision is the ratio of correctly predicted positive observations to the total predicted positives. Recall (Sensitivity) is the ratio of correctly predicted positive observations to the all observations in actual class. The F1 Score is the weighted average of Precision and Recall.
- **Area Under the ROC Curve (AUC-ROC):** A plot of the true positive rate against the false positive rate, summarizing the model's performance across all classification thresholds.
- **Confusion Matrix:** A table showing the number of correct and incorrect predictions categorized by class.

Model Evaluation

Regression Problems:

- **Mean Absolute Error (MAE):** The average of the absolute differences between the predicted and actual values.
- **Mean Squared Error (MSE):** The average of the squared differences between the predicted and actual values.
- **Root Mean Squared Error (RMSE):** The square root of the MSE.
- **R-Squared:** Indicates the proportion of variance in the dependent variable that is predictable from the independent variables.

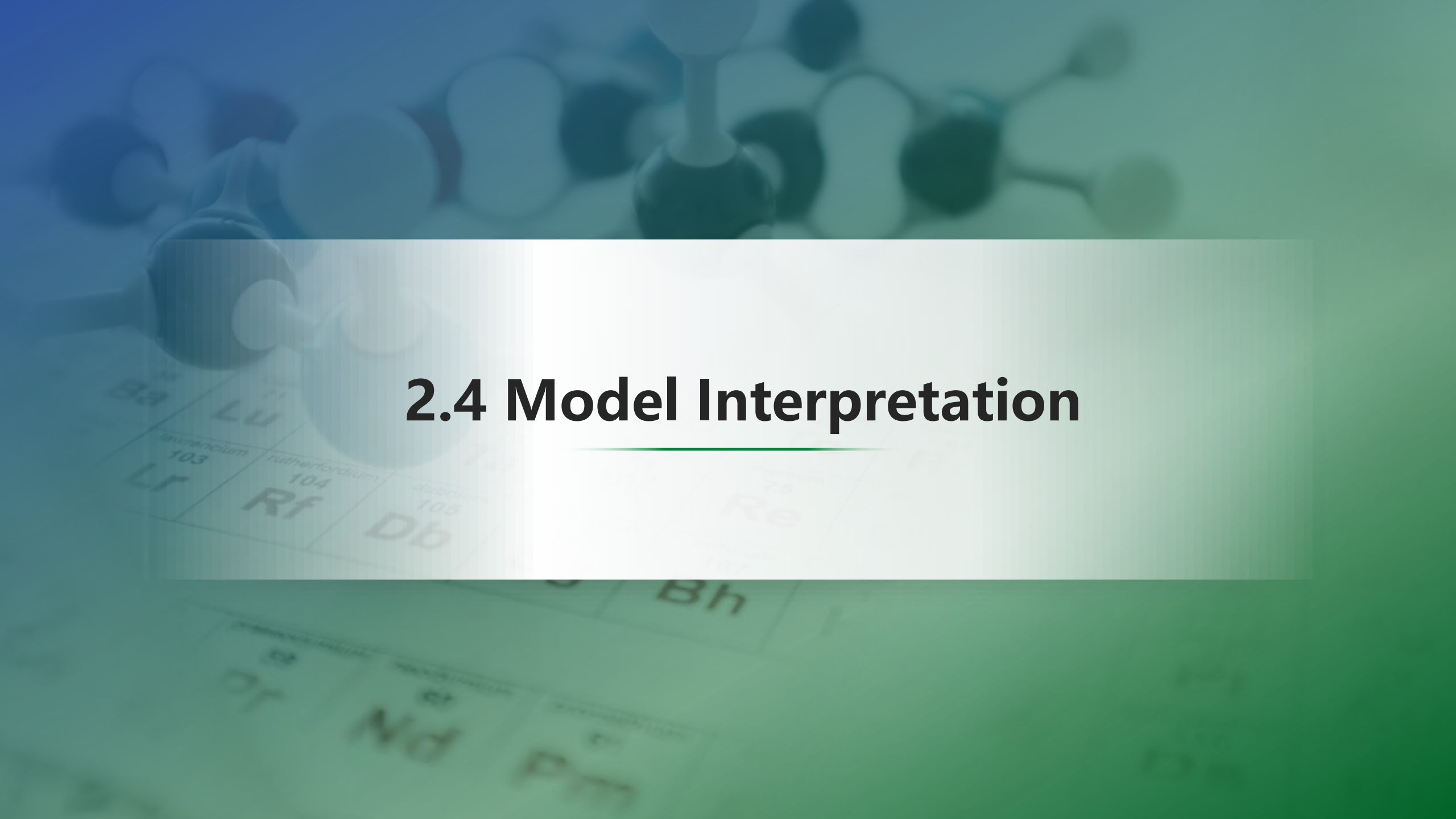
Model Evaluation

General Metrics:

- **Log-Loss:** Suitable for binary and multiclass classification problems. It provides a measure of uncertainty.
- **Brier Score:** Measures the mean squared difference between predicted probabilities and the actual outcomes.

Comparative Metrics:

- **Model Comparison:** Comparing different models' metrics to choose the one that performs best on the validation or test data.
- **Statistical Significance Testing:** Performing statistical tests like the paired t-test to compare the performance of different models.

The background of the slide features a blurred molecular model with white and grey spheres connected by rods, and a portion of a periodic table. The periodic table shows elements like Lu, Rf, Db, Re, Bh, Nd, and Pt. A semi-transparent white box is centered over the image, containing the section title.

2.4 Model Interpretation

Model Interpretation

Feature importance: techniques that assign a score to input features based on how useful they are at predicting a target variable.

Key Methods

Permutation Importance: "Measures changes in model accuracy when a feature's values are randomly shuffled."

Feature Importance from Model Coefficients: "Utilizes coefficients in linear models as importance scores."

Tree-based Methods: "E.g., Gini importance from Random Forests, based on the decrease in node impurity."

Applications and Benefits

- Helps in feature selection and reducing model complexity
- Enhances model interpretability for stakeholders
- Identifies potential areas of research or data collection

Considerations

- "Dependence on the chosen model and data."
- "Potential bias in importance scores."
- "Importance does not imply causality."

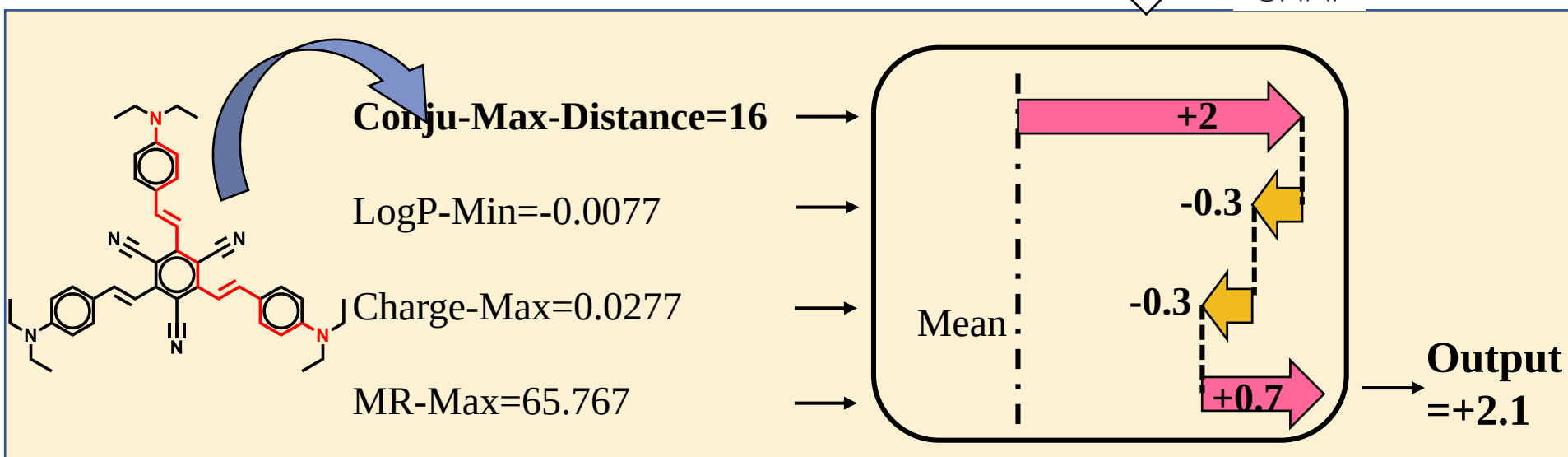
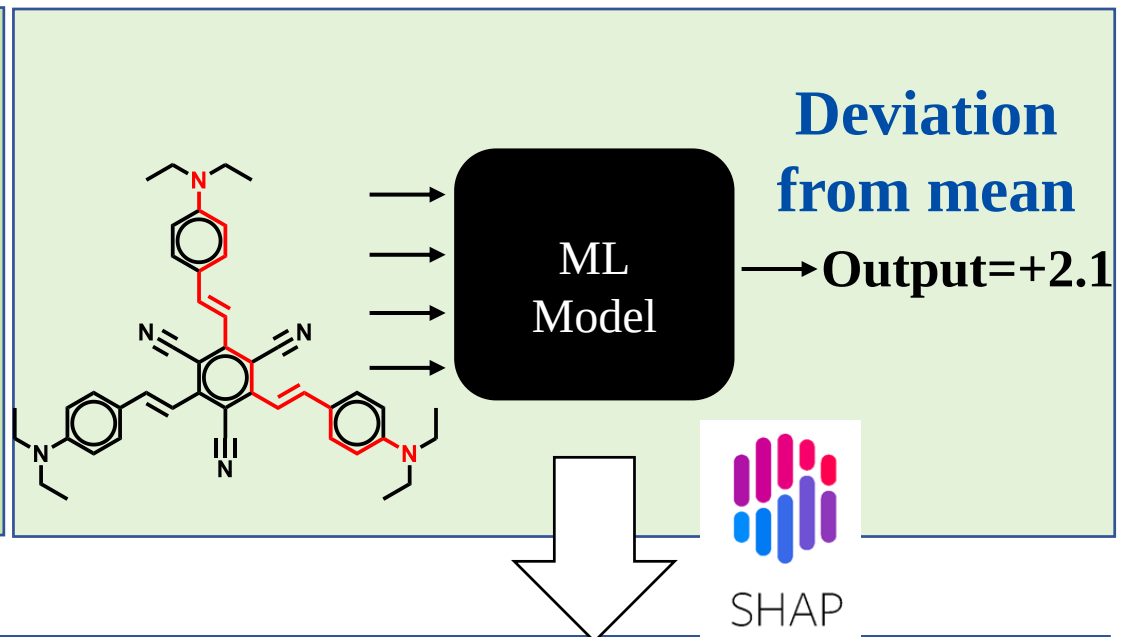
Isolating Contribution: Shapley Value

SHapley Additive exPlanation

Based on **Game Theory**

Isolating Contribution
from a single feature

Additive
Explainer



Other Methods for Model Interpretation

LIME (Local Interpretable Model-agnostic Explanations):

- LIME provides explanations for individual predictions by approximating the local decision boundary around a given prediction. It works by generating new samples around the instance and observing the changes in predictions.

Partial Dependence Plots (PDPs):

- PDPs illustrate the effect of a single or two features on the predicted outcome, keeping all other features constant. This helps in understanding the relationship between the feature(s) and the prediction.

Global Surrogate Models:

- This approach involves training a simpler, more interpretable model (like a decision tree or linear model) to approximate the predictions of a complex model. The surrogate model is then analyzed to understand the more complex model.

Counterfactual Explanations:

- Counterfactual explanations provide insights by illustrating how altering certain features could change the prediction to a desired outcome. It answers "what-if" scenarios, offering actionable insights.

Rule-based Methods

Rule-Based Models: Models like rule lists or sets provide clear, if-then rules for predictions, making them inherently interpretable.

Anchors: Anchors provide rule-based explanations for individual predictions, highlighting the features that were critical in making a specific decision.

Explicit Rules: Rule-based systems operate on a series of if-then statements. These rules are explicitly defined and often rely on domain expertise. For example, a rule in a medical diagnosis system might be "if fever is high and cough is persistent, then consider a diagnosis of pneumonia."

Transparency and Interpretability: One of the advantages of rule-based models is their transparency. The decision-making process is clear and understandable because it follows the predefined rules.

Limited Flexibility: These models can be less flexible and might not adapt well to complex or nuanced scenarios that haven't been explicitly programmed into the rules.

Domain-Specific: They are often highly tailored to specific domains, requiring significant expertise to develop accurate and comprehensive rules.

Maintenance: Rule-based systems can be easier to maintain and debug due to their transparency, but updating them requires manual rule adjustment.

Attention

Attention mechanisms in machine learning, particularly in the context of neural networks, have become a critical component for improving the performance and interpretability of models.

Focus on Relevant Information: Attention mechanisms allow a model to focus on specific parts of the input data that are more relevant for making a decision or prediction.

Dynamic Weighting: In a neural network, attention dynamically assigns weights to different parts of the input. For example, in a sentence, certain words may be given more importance for understanding the overall meaning depending on the context.

Types of Attention Mechanisms

1.Soft Attention: Assigns weights in a differentiable way, allowing for end-to-end training of the model using standard backpropagation.

2.Hard Attention: Selects certain parts of the input to focus on and ignores others, often non-differentiable and requiring special training techniques like reinforcement learning.

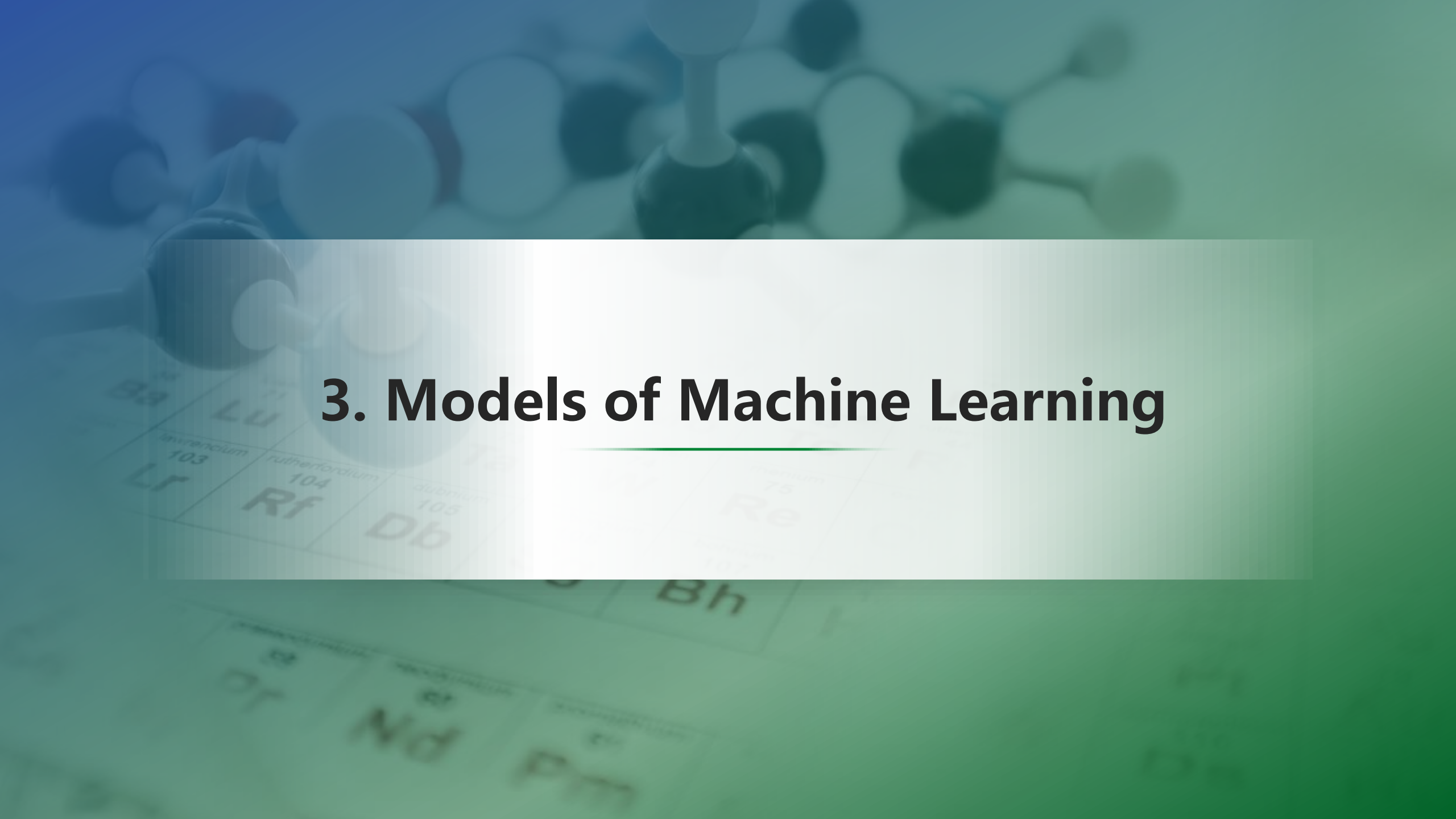
Attention

Attention in Model Interpretability

- 1. Transparency of Decision-Making:** Attention mechanisms can make neural networks more interpretable. By examining the attention weights, we can get a sense of which parts of the input data the model is focusing on when making a decision.
- 2. Visualizations:** In tasks like text translation or image captioning, attention weights can be visualized to show which words in the source text are being focused on for each word in the translated text, or which parts of an image are being focused on while generating a caption.
- 3. Improved Diagnostics:** Attention can help in diagnosing model performance. For instance, if a model makes an incorrect prediction, examining the attention weights can reveal whether it was focusing on irrelevant or incorrect parts of the input.

Challenges and Limitations

- 1. Not Always Intuitive:** Attention weights do not always align with human intuition. High attention weights on certain inputs do not necessarily mean those inputs are more 'important' in a human-understandable way.
- 2. Complex Models:** In very deep or complex models, interpreting attention can become challenging, as the interaction of layers and attention mechanisms can be intricate.
- 3. Correlation vs. Causation:** Attention can show correlation (what the model is looking at) but not causation (why looking at that part helps in decision-making).



3. Models of Machine Learning

Models of ML

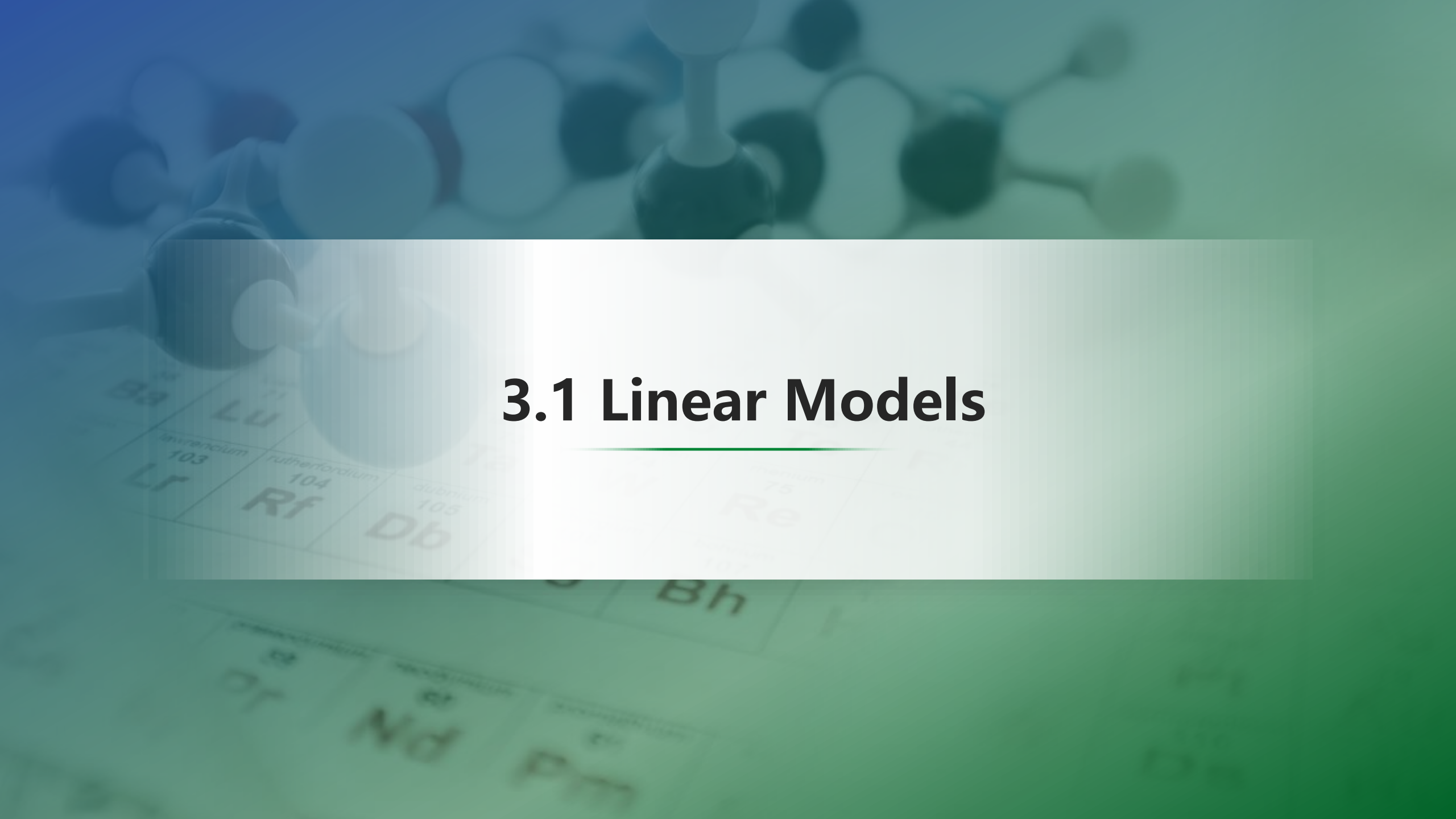
3.1 Linear Model

3.2 Decision Tree

3.3 Support Vector Machine

3.4 Bayesian Methods

3.5 Ensemble Models

The background of the slide features a blurred molecular model with white and grey spheres connected by rods, and a portion of the periodic table. The periodic table shows elements like Lu (Lutetium), Rf (Rutherfordium), Db (Dubnium), Re (Rhenium), Bh (Bohrium), and Nd (Neodymium).

3.1 Linear Models

Linear Regression Model

$$f(\mathbf{x}) = w_1 x_1 + w_2 x_2 + \dots + w_d x_d + b$$

↓ ↓ ↓
sample weight feature

Vector Form

$$f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b \quad \mathbf{w} = (w_1; w_2; \dots; w_d)$$

Is it Moutai

$$f_{IsMoutai}(\mathbf{x}) = 0.2 x_{color} + 0.5 x_{taste} + 0.3 x_{smell} + 1$$

The model can also be used for classification

e.g. Binary Classification

$$z = \mathbf{w}^T \mathbf{x} + b \quad y \in \{0, 1\} \quad \text{class 0 and class 1}$$

Need a function to project z to y

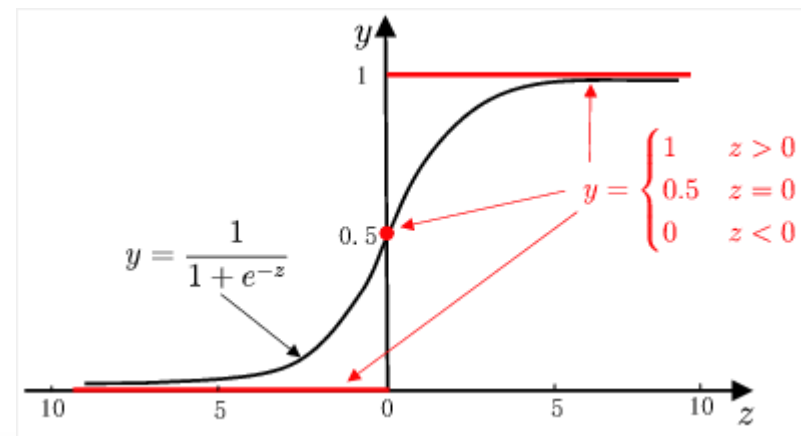
➤ Unit step function

$$y = \begin{cases} 0, & z < 0; \\ 0.5, & z = 0; \\ 1, & z > 0, \end{cases} \quad \text{Discontinuous}$$

➤ Logistic function

$$y = \frac{1}{1 + e^{-z}}$$

Differentiable



Logistic regression

$$\ln \frac{p(y = 1 | \mathbf{x})}{p(y = 0 | \mathbf{x})} = \mathbf{w}^T \mathbf{x} + b$$

$$p(y = 1 | \mathbf{x}) = \frac{e^{\mathbf{w}^T \mathbf{x} + b}}{1 + e^{\mathbf{w}^T \mathbf{x} + b}}$$

$$p(y = 0 | \mathbf{x}) = \frac{1}{1 + e^{\mathbf{w}^T \mathbf{x} + b}}$$

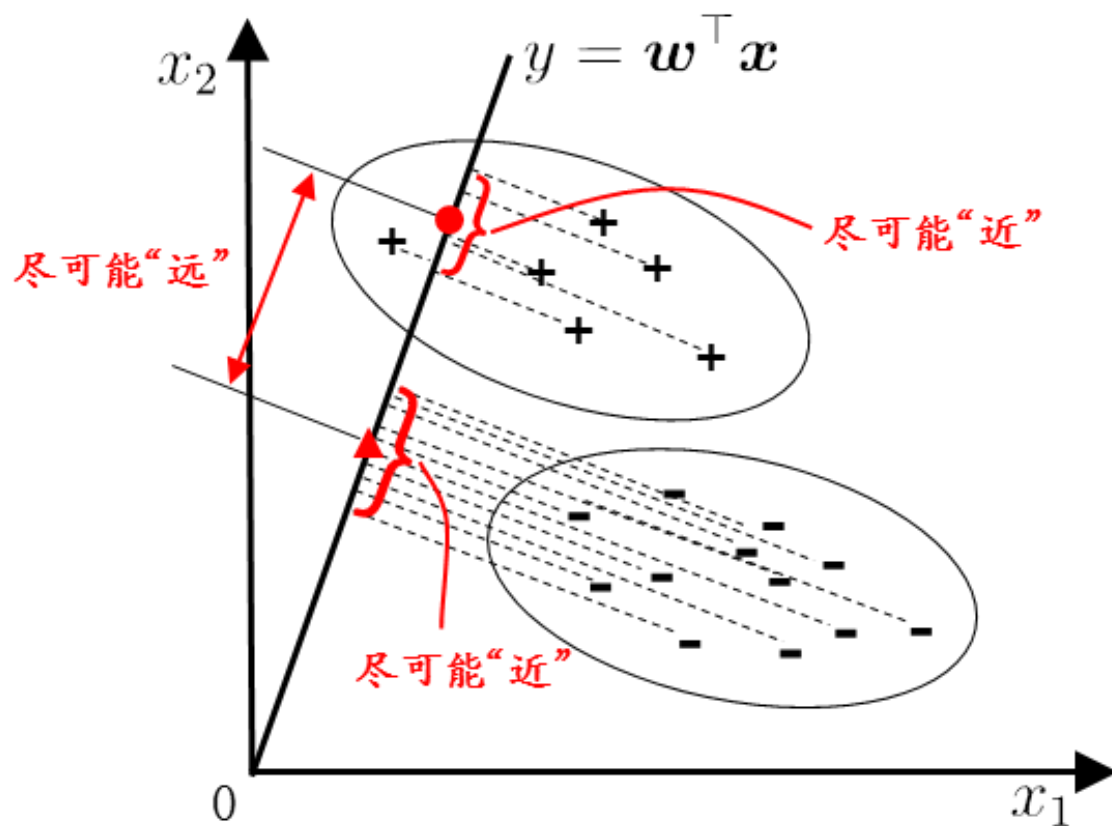
Using all the techniques of linear regression to solve binary classification problem

Getting the probability of positive or negative

$$f_{IsMoutai}(\mathbf{x}) = 0.2 x_{color} + 0.5 x_{taste} + 0.3 x_{smell} + 1$$

$$p(y = 1 | \mathbf{x}) = f / (f + 1)$$

Linear Discriminant Analysis



LDA is also a technique for dimensional reduction

Principle: projecting to a line that let the points in the same cluster as close as possible to each other, while centers of different clusters as far away as possible

Multi-class Classification

Use binary classification machine to solve multi-class problem

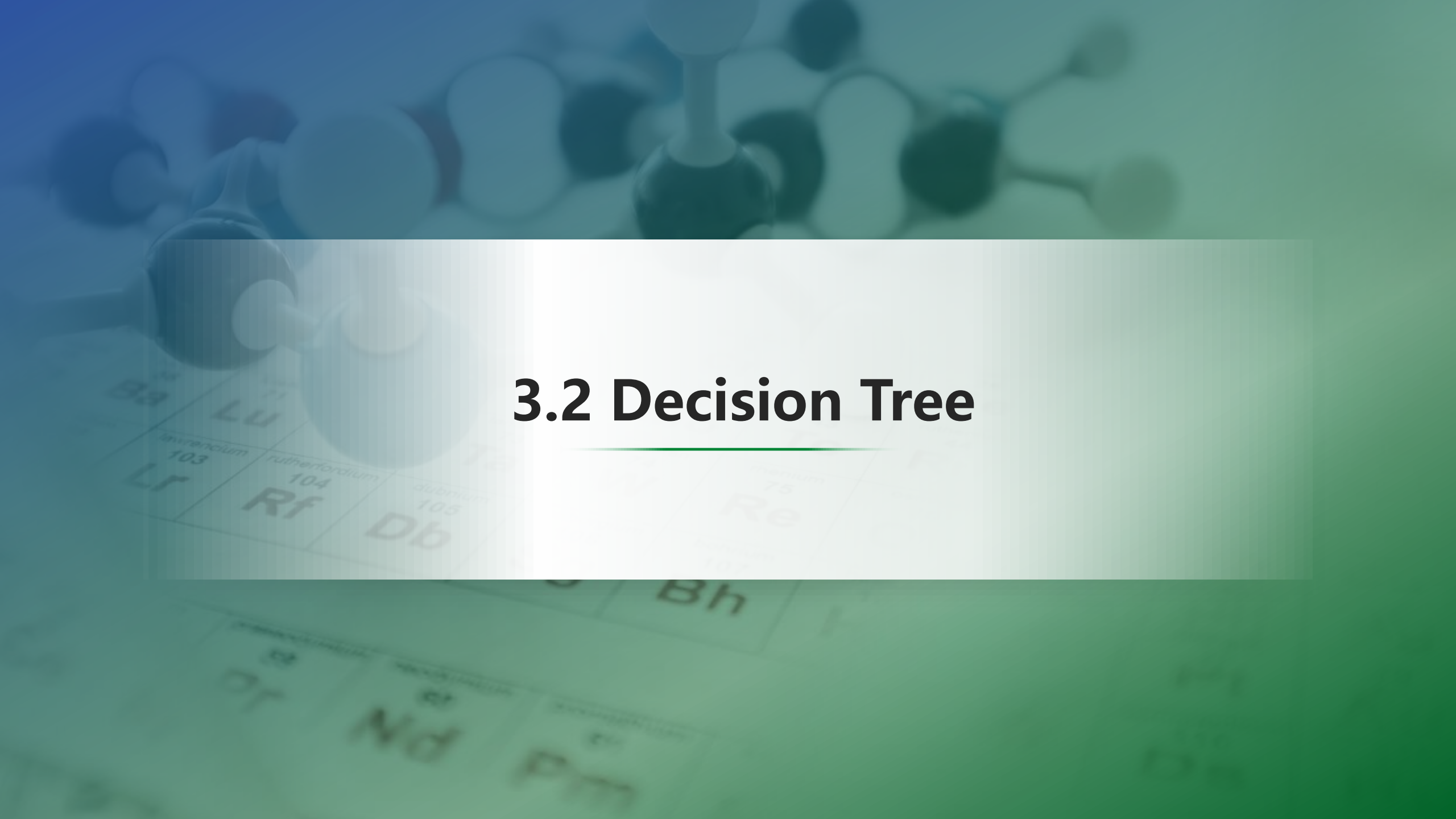
- One vs. One, OvO
- One vs. Rest, OvR
- Many vs. Many, MvM

Class imbalance

If the number of positive samples and the number of negative samples have a huge difference.

- **Undersampling:** removing samples to achieve a balance
e.g. EasyEnsemble
- **Oversampling:** adding some fake samples (based on label of neighbors)
e.g. SMOTE
- **Threshold-moving**

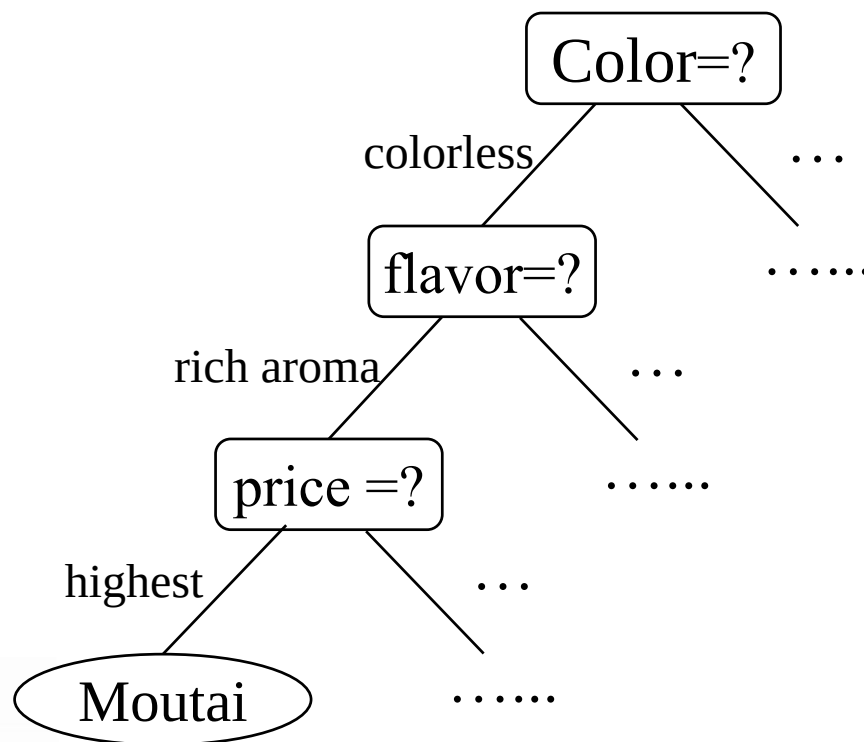
$$\frac{y}{1-y} > 1 \Rightarrow \frac{y}{1-y} > \frac{m^+}{m^-} \quad \text{positive / negative sample ratio}$$



3.2 Decision Tree

Decision Tree

Is it Moutai?



How to decide which is the best feature to choose?

Increase node “purity”

- Information Gain
- Gain Ratio
- Gini Index

Information Entropy (C4.5; ID3)

$$\text{Ent}(D) = - \sum_{k=1}^{|\mathcal{Y}|} p_k \log_2 p_k \quad p_k \text{ is the fraction of the } k\text{'th class}$$

We define if $p = 0$ then $p \log_2 p = 0$

After we select each feature a in the tree, the information gain can be calculated

$$\text{Gain}(D, a) = \text{Ent}(D) - \sum_{v=1}^V \frac{|D^v|}{|D|} \text{Ent}(D^v)$$

Gini Index (CART)

$$\text{Gini}(D) = \sum_{k=1}^{|\mathcal{Y}|} \sum_{k' \neq k} p_k p_{k'} = 1 - \sum_{k=1}^{|\mathcal{Y}|} p_k^2$$

The smaller the gini index, the better the classification

Pruning

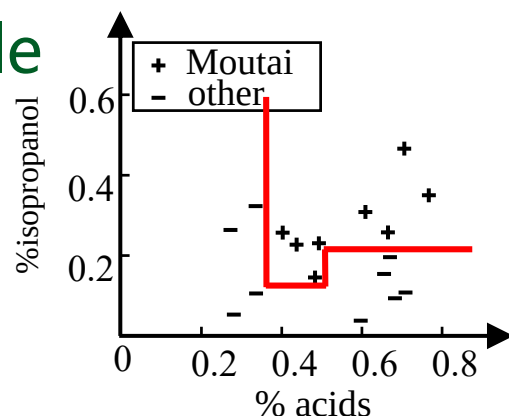
- To simplify the model to avoid overfitting
- Not commonly used in ensemble tree methods

Regression

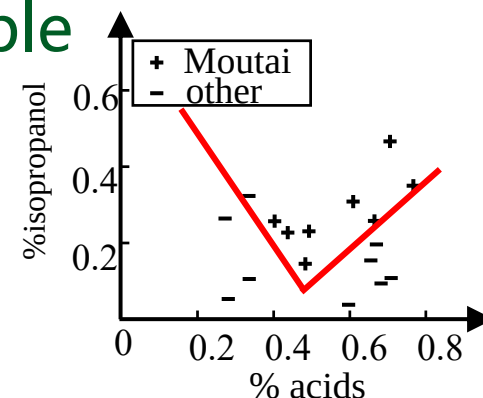
Decision tree can also be used for regression. Just treat the continuous value as a distribution on multiple classes

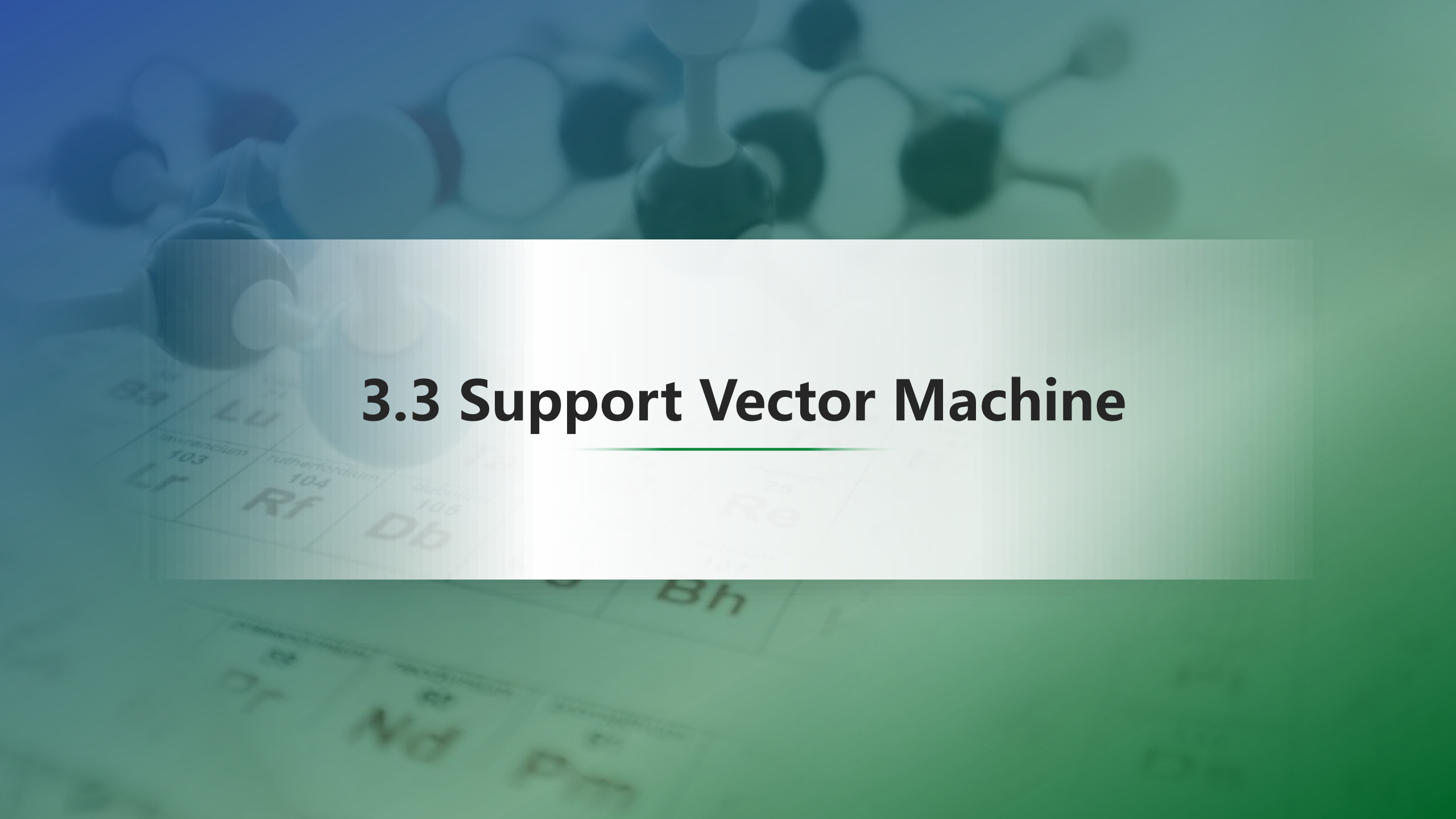
Multi-variable decision tree

uni-variable



multi-variable



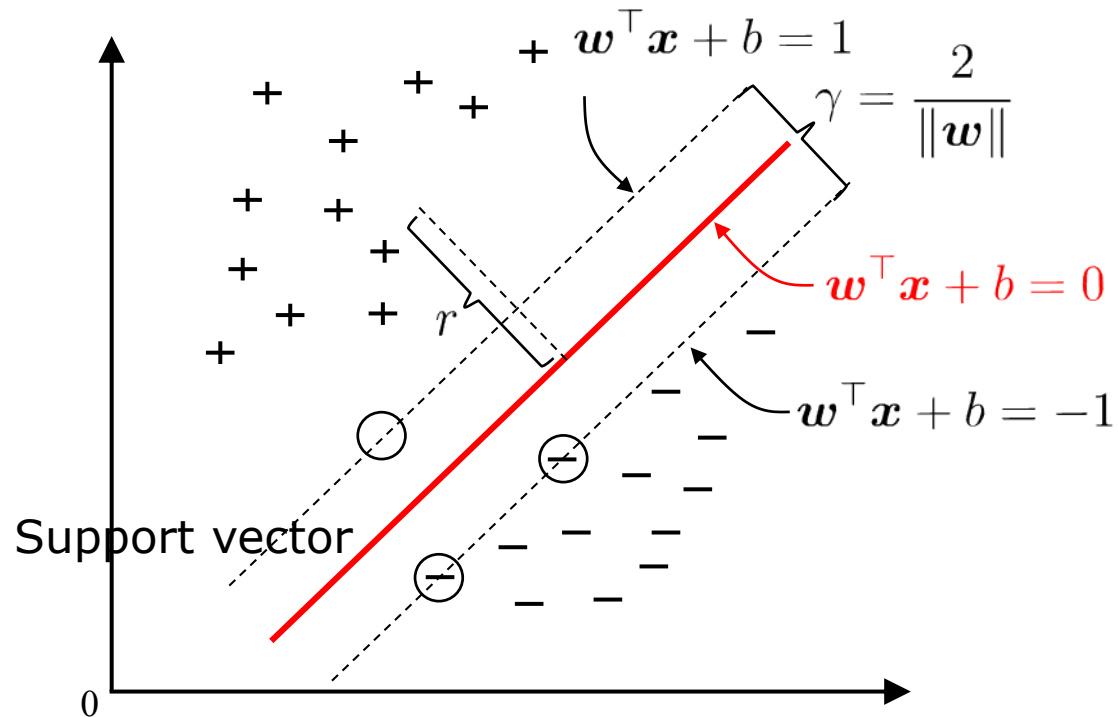


3.3 Support Vector Machine

Support Vector Machine

Linear Model: finding a hyperplane to separate different samples

Hyperplane equation $\mathbf{w}^\top \mathbf{x} + b = 0$



$$\begin{aligned} \arg \max_{\mathbf{w}, b} \quad & \frac{2}{\|\mathbf{w}\|} \\ \text{s.t.} \quad & y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1, \quad i = 1, 2, \dots, m. \end{aligned}$$



$$\begin{aligned} \arg \min_{\mathbf{w}, b} \quad & \frac{1}{2} \|\mathbf{w}\|^2 \\ \text{s.t.} \quad & y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1, \quad i = 1, 2, \dots, m. \end{aligned}$$

Kernel Support Vector Machine

Kernel: nonlinear mapping to a linear space using kernel function

$$f(\mathbf{x}) = \mathbf{w}^\top \phi(\mathbf{x}) + b$$

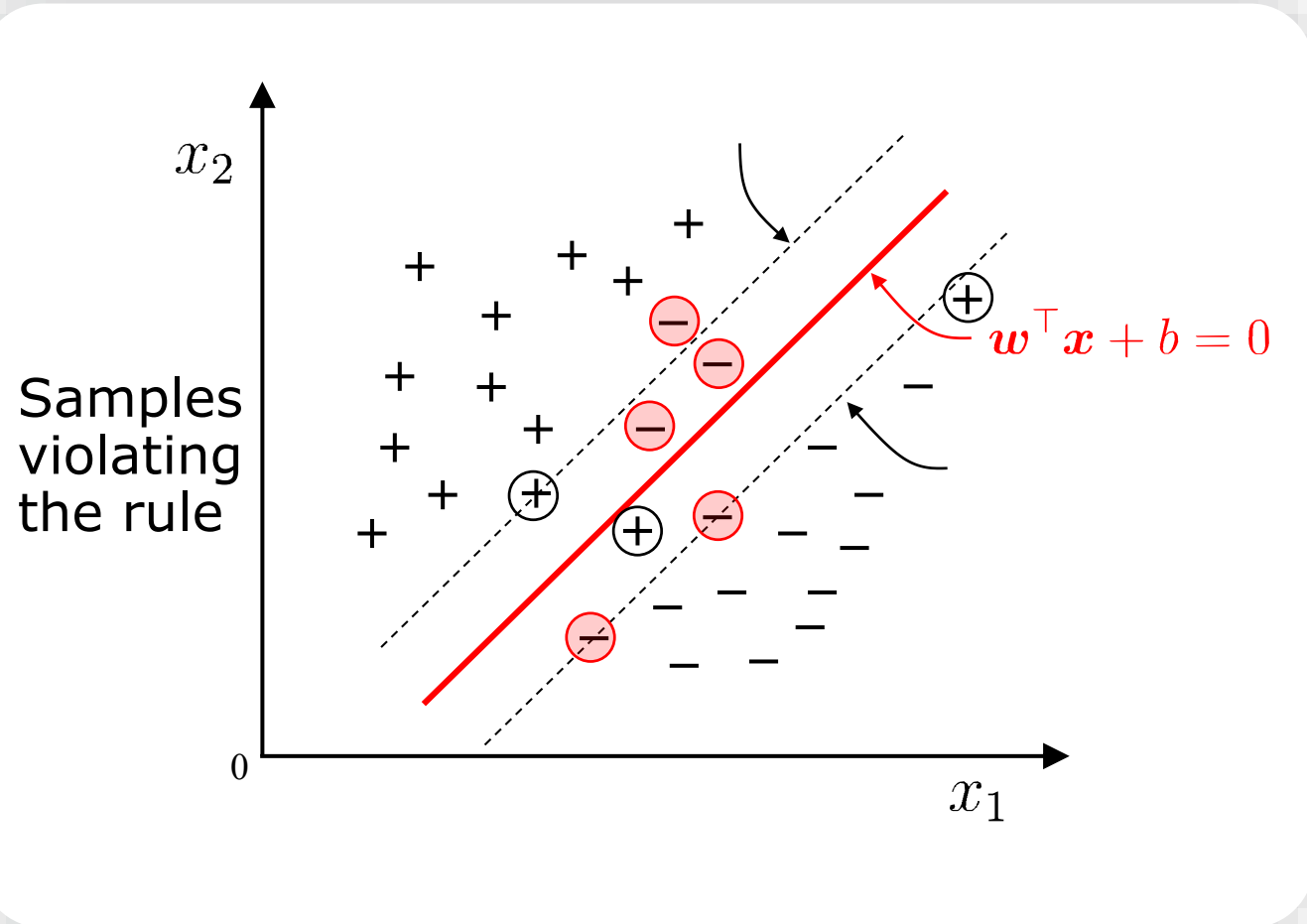
$$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^\top \phi(\mathbf{x}_j)$$

Common kernel functions Kernel Method can be applied to other linear models

名称	表达式	参数
线性核	$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i^\top \mathbf{x}_j$	
多项式核	$\kappa(\mathbf{x}_i, \mathbf{x}_j) = (\mathbf{x}_i^\top \mathbf{x}_j)^d$	$d \geq 1$ 为多项式的次数
高斯核	$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\ \mathbf{x}_i - \mathbf{x}_j\ ^2}{2\delta^2}\right)$	$\delta > 0$ 为高斯核的带宽(width)
拉普拉斯核	$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\ \mathbf{x}_i - \mathbf{x}_j\ }{\delta}\right)$	$\delta > 0$
Sigmoid核	$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \tanh(\beta \mathbf{x}_i^\top \mathbf{x}_j + \theta)$	$\tanh$ 为双曲正切函数, $\beta > 0, \theta < 0$

Soft Support Vector Machine

Allow some samples to violate the rule

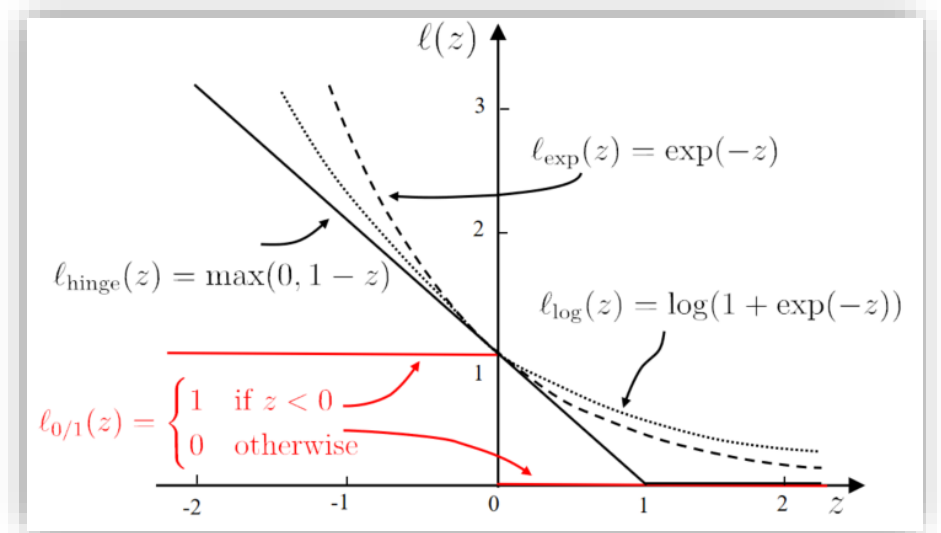


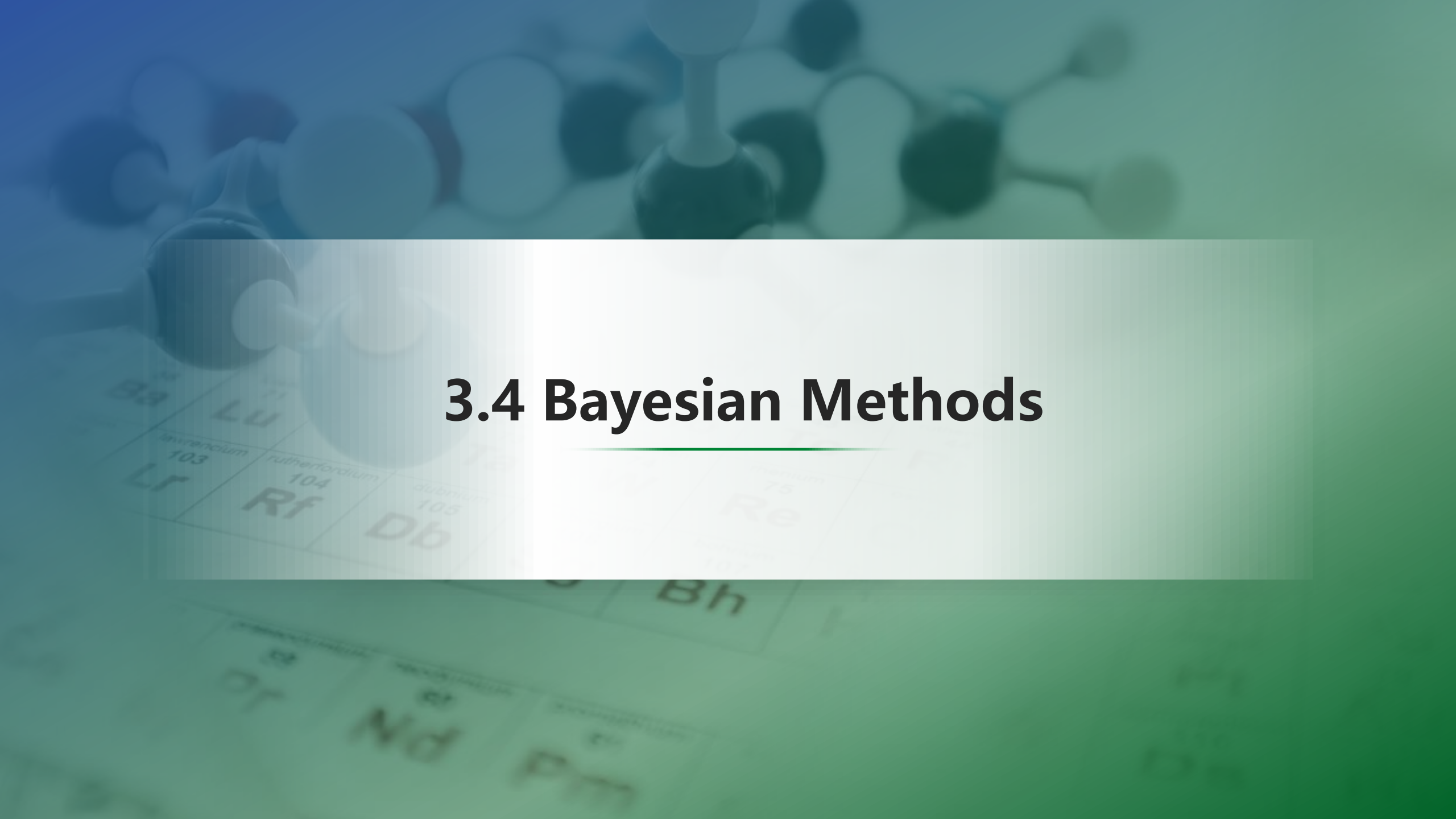
$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^m l_{0/1} (y_i (\mathbf{w}^T \phi(\mathbf{x}_i) + b) - 1)$$

$$l_{0/1} = \begin{cases} 1 & z < 0 \\ 0 & \text{otherwise} \end{cases}$$

Non-convex, discontinuous

substitution





3.4 Bayesian Methods

Bayesian decision theory

There are N possible categories $y = \{c_1, c_2, \dots, c_N\}$

λ_{ij} is the loss in mistaking c_j as c_i

Conditional risk:

$$R(c_i | \mathbf{x}) = \sum_{j=1}^N \lambda_{ij} P(c_j | \mathbf{x}) \quad \lambda_{i,j} \begin{cases} 0, & \text{if } i = j; \\ 1, & \text{otherwise,} \end{cases}$$

We are searching for a rule $h : X \mapsto Y$

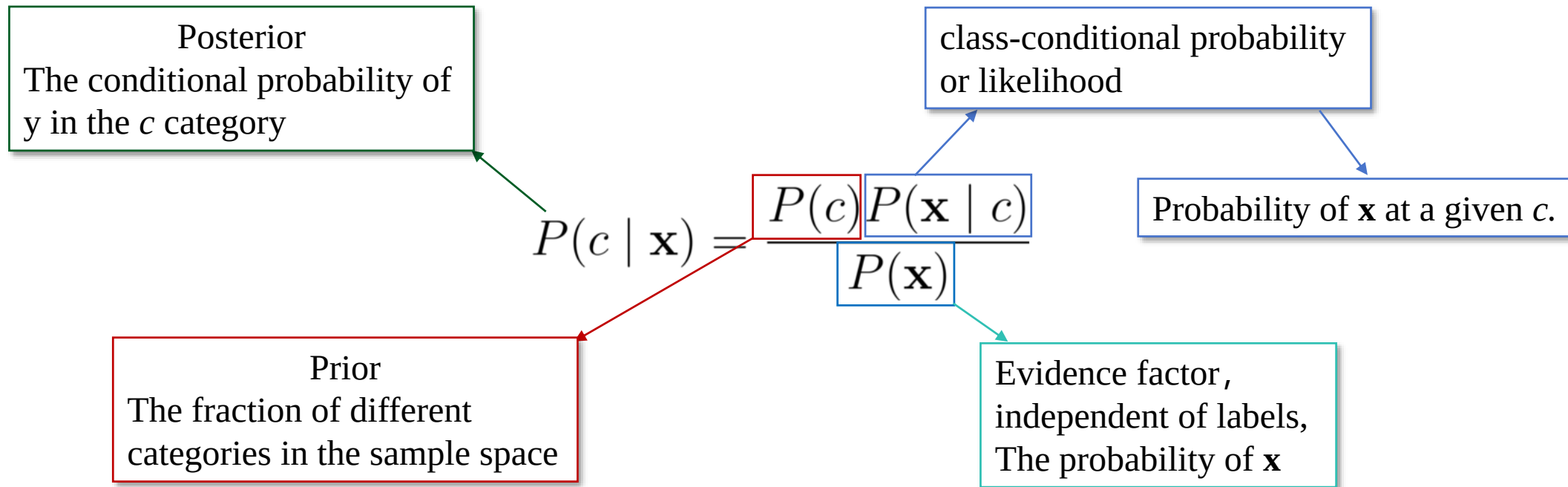
to minimize $R(h) = \mathbf{E}_x [R(h(\mathbf{x}) | \mathbf{x})]$

An Bayes optimal classifier is the one that can minimize risk for every sample

$$h^*(x) = \operatorname{argmin}_{c \in y} R(c | x)$$

$R(h^*)$ is Bayes risk

Bayes Theorem



$P(\mathbf{x} | c)$ The key is to obtain the likelihood

Frequentist Approach

- Probabilities are related to frequencies of events in repeated experiments
- Maximum Likelihood Estimation (MLE) finds the parameter values that maximize the likelihood function, given the observed data.
- Frequentist methods do not incorporate prior beliefs or information into the analysis.

vs.

Bayesian Approach

- Probability is interpreted as a degree of belief or a measure of the state of knowledge about events, including unobservable parameters.
- Bayesian inference uses Bayes' Theorem, which involves updating prior beliefs with the likelihood of the observed data to produce a posterior probability distribution.
- Bayesian methods are inherently subjective, as they depend on the prior distribution chosen by the analyst. They are flexible in incorporating prior information and are particularly useful in situations where data is limited or prior knowledge is strong.

Naïve Bayes Classifier

attribute conditional independence assumption

The features are independent of each other

$$P(c \mid \mathbf{x}) = \frac{P(c)P(\mathbf{x} \mid c)}{P(\mathbf{x})} = \frac{P(c)}{P(\mathbf{x})} \prod_{i=1}^d P(x_i \mid c)$$

$$h_{nb}(\mathbf{x}) = \operatorname{argmax}_{c \in y} P(c) \prod_{i=1}^d P(x_i \mid c)$$

Naïve Bayes Classifier

Estimate Prior $P(c)$

D_c is a subset of the training set D
Belonging to category c

$$P(c) = \frac{|D_c|}{D}$$

Estimate Conditional Probability $P(x_i | c)$

D_{c,x_i} is a subset in D_c with x_i value

For discrete feature:

$$P(x_i | c) = \frac{|D_{c,x_i}|}{D_c}$$

For continuous feature:

$$P(x_i | c) = \frac{1}{\sqrt{2\pi}\sigma_{c,i}} \exp\left(-\frac{(x_i - \mu_{c,i})^2}{2\sigma_{c,i}^2}\right)$$

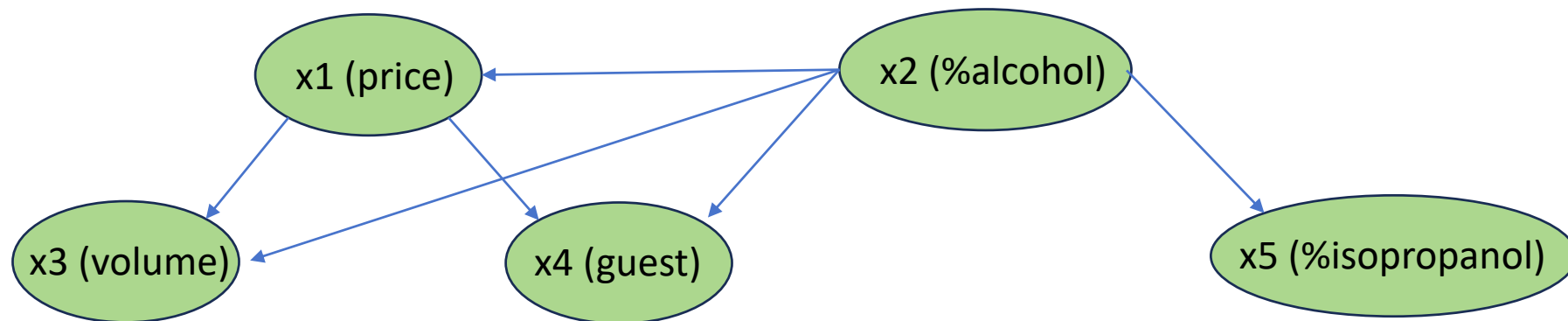
Bayes network

also known as **a belief network**

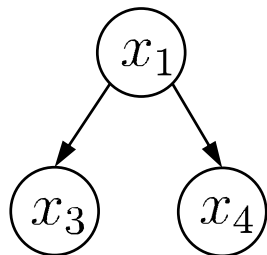
a probabilistic graphical model that represents a set of variables and their conditional dependencies via a directed acyclic graph (DAG)

- **Nodes:** Each node represents a variable, which can be either observed data or unknown parameters. In the context of chemistry or material science, these variables could be properties of a substance, experimental conditions, or outcomes.
- **Directed Edges:** Arrows connect pairs of nodes, indicating the direction of dependence. For example, an arrow from node A to node B implies that A has a direct influence on B.
- **Conditional Probability Tables (CPTs):** For each node, a CPT specifies the probability of the node taking on each of its possible values, given the values of its parent nodes. For nodes without parents (root nodes), the CPT specifies the marginal probability of the node's values.

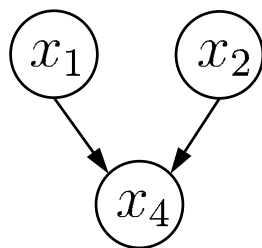
Bayes network



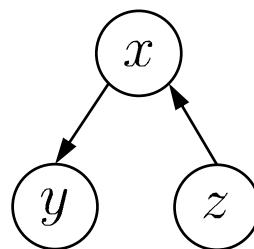
$$P(x_1, x_2, x_3, x_4, x_5) = P(x_1|x_2) P(x_2) P(x_3|x_1, x_2) P(x_4|x_1, x_2) P(x_5|x_2)$$



同父结构



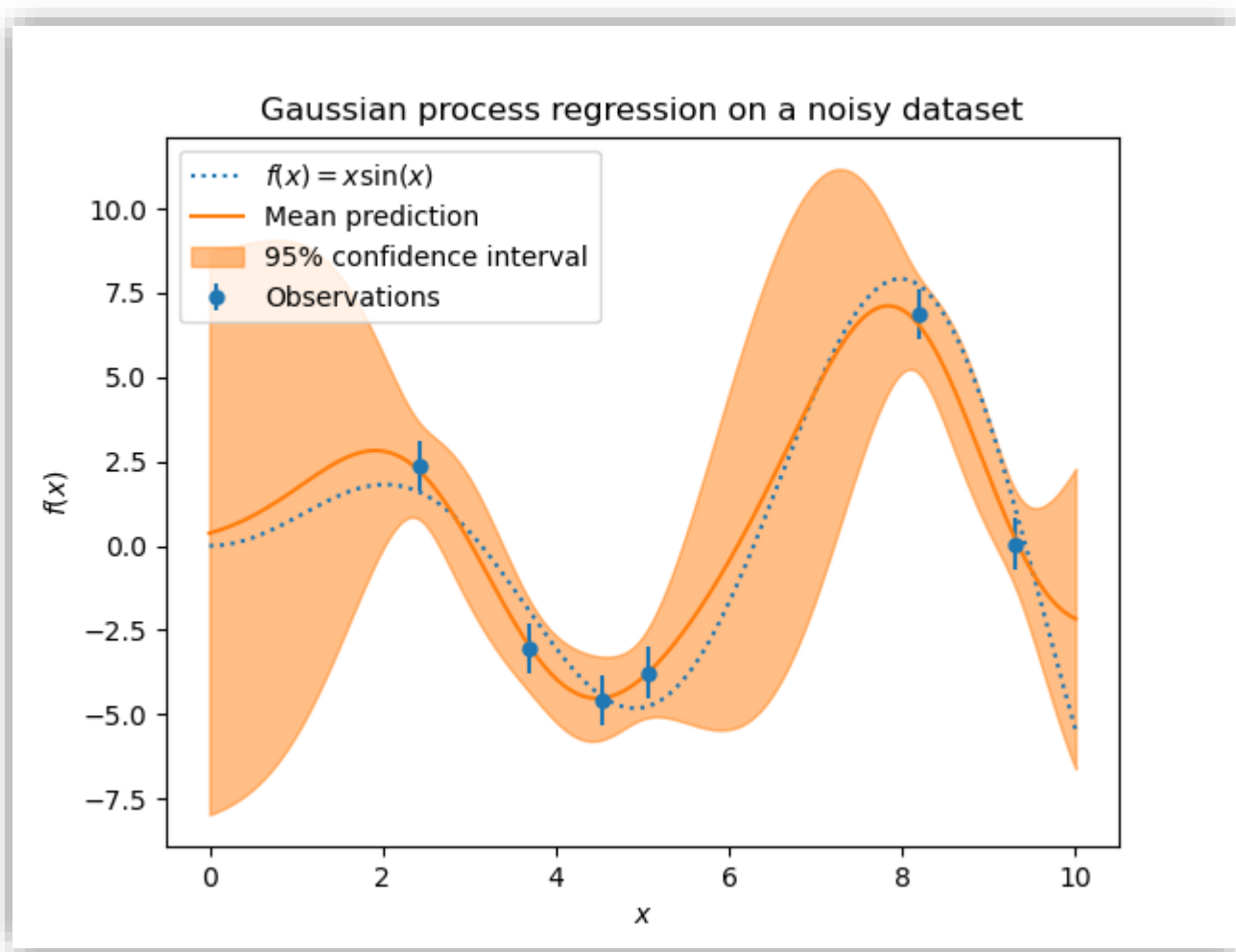
V型结构



顺序结构

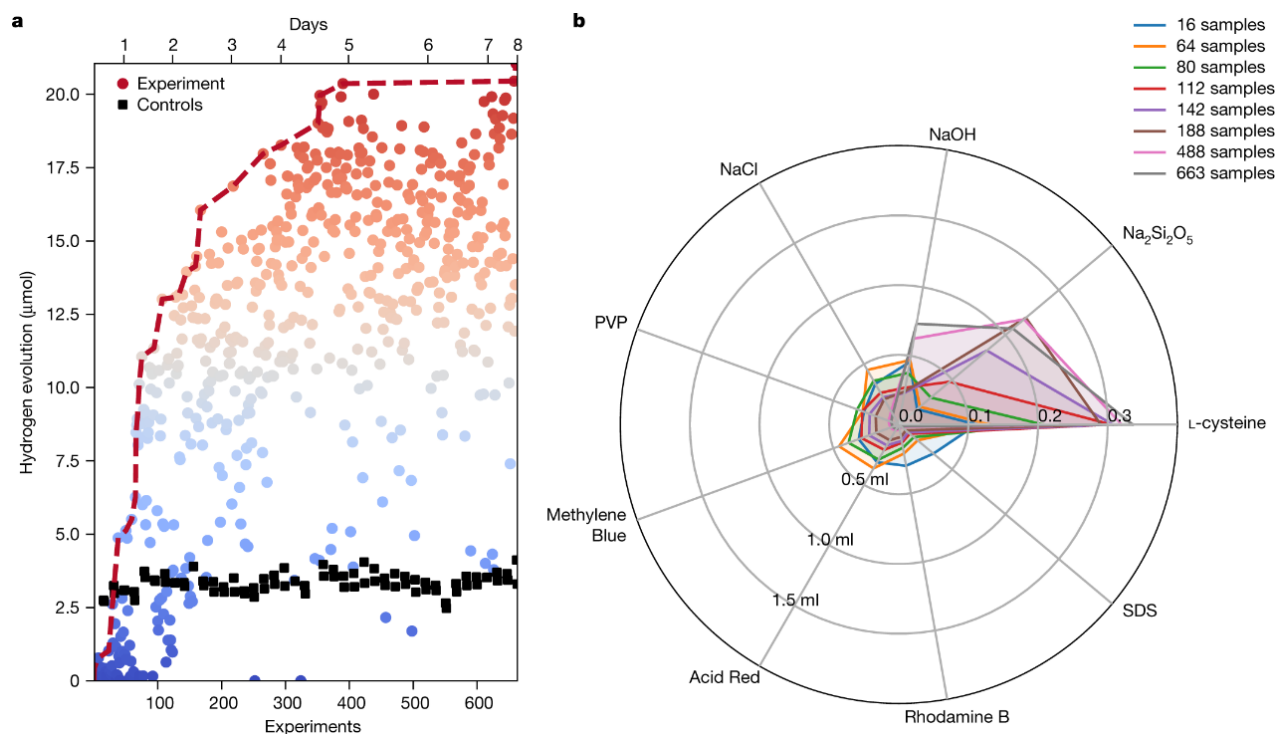
Gaussian Process

- Using kernel function to specify covariance
- Can consider error in the data

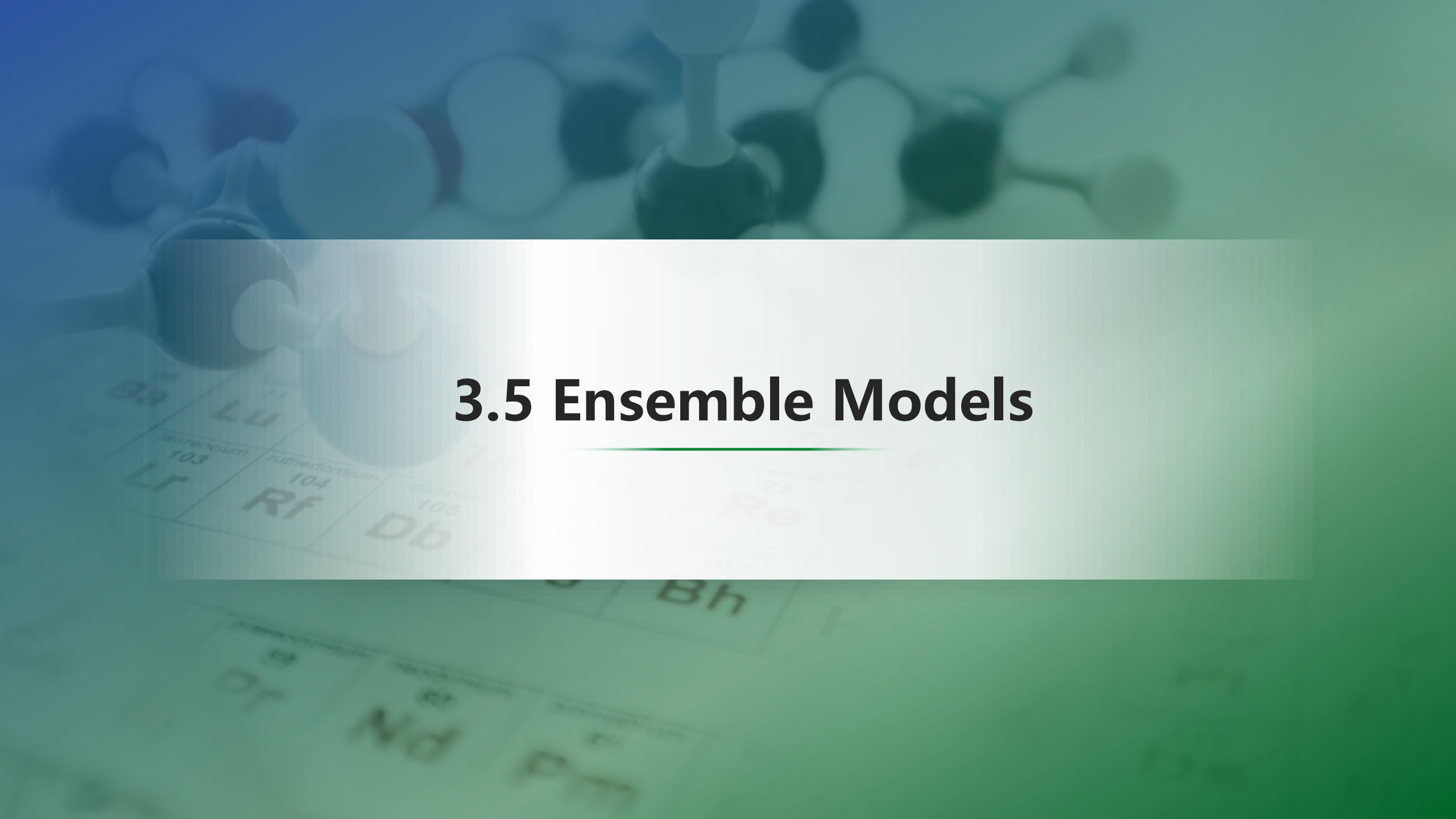


Bayesian Optimization

Bayesian optimization of photocatalysis

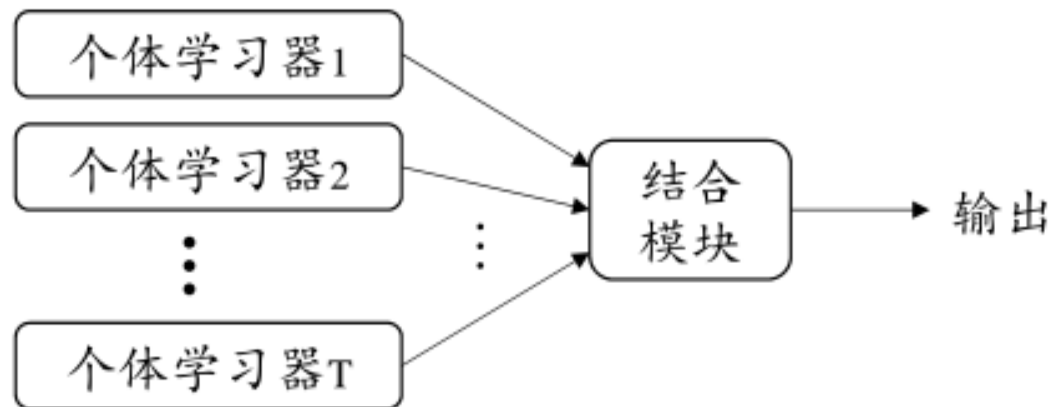


Bayesian Optimization is a strategy for the global optimization of black-box functions that are expensive to evaluate. Bayesian optimization builds a probabilistic model of the function using Bayes theorem and uses it to select the most promising candidates for evaluation.

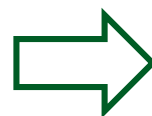


3.5 Ensemble Models

Ensemble learning



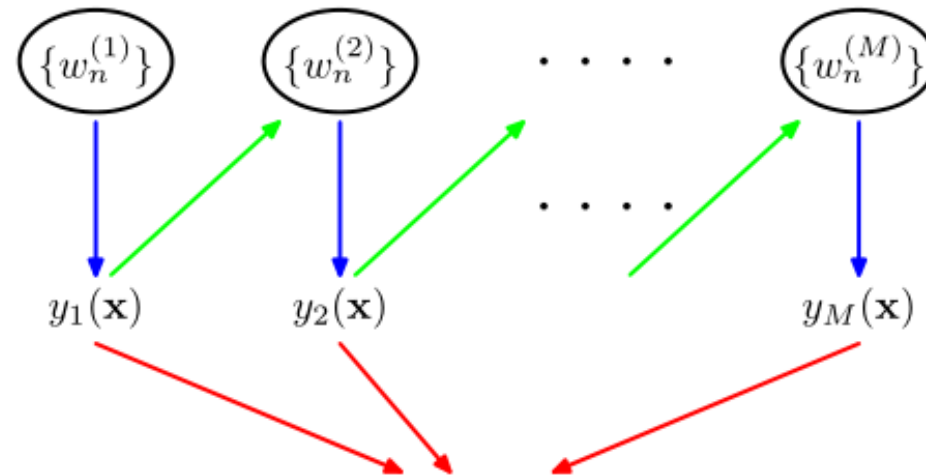
Good and different
Diversity matters



Challenge: good models
cannot be independent and
too different

Boosting

- Individual learners are highly dependent on each other
- Update data distribution each time
- Use all learners for output



AdaBoost Methods

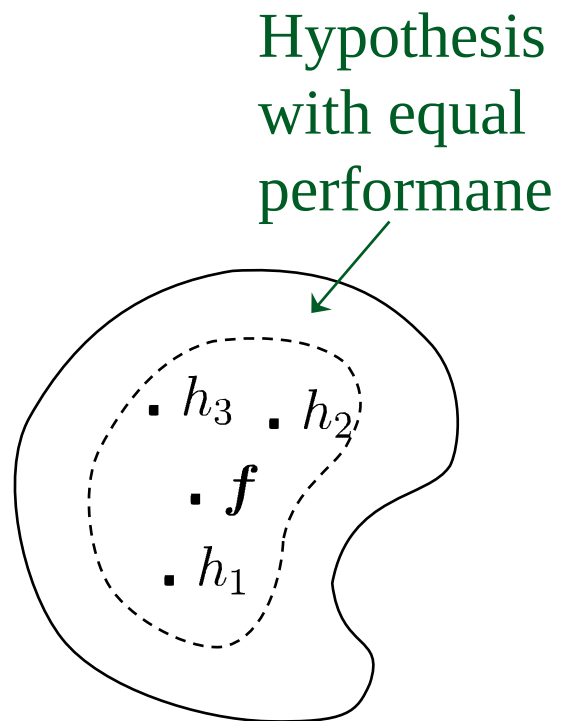
$$Y_M(\mathbf{x}) = \text{sign} \left(\sum_m^M \alpha_m y_m(\mathbf{x}) \right)$$

Bagging

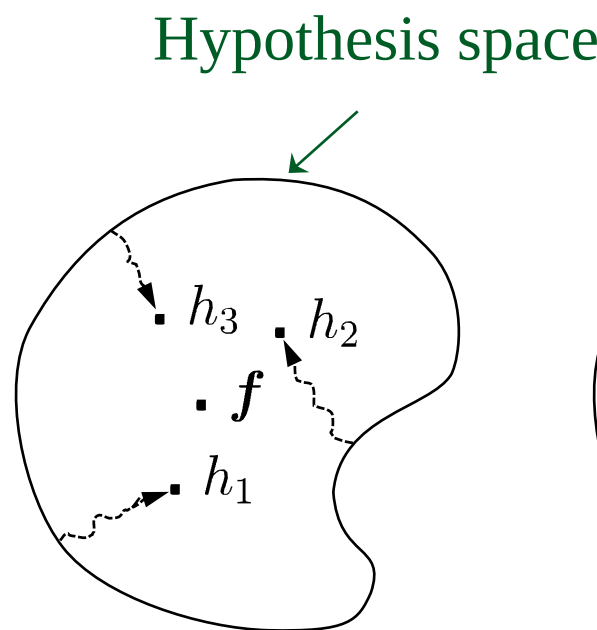
- Individual learners are independent
 - Parallel in training
 - Bootstrap sampling
-
- Efficient in time
 - Use out-of-bag data points for assessment without the need of a validation set

Random Forest is a modified version of Bagging

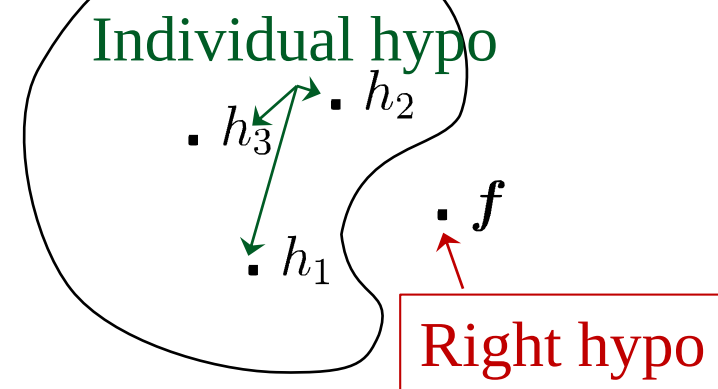
Why ensemble can improve performance ?



(a) Statistics



(b) Calculation



(c) Representation